Nios V ↔ Caliptra 1.x Integration Note

- **Project:** Caliptra 1.x (passive) on Intel Agilex 5 E **A5EC065BB32AE6S**, Nios V/m mailbox host (AXI4-lite → Avalon → APB).
- **Repo:** pwilliam282/caliptra-a5e-quartus
- **This document:** `docs/NIOS_CALIPTRA_INTEGRATION.md`
- **RTL / spec version:** chipsalliance/caliptra-rtl **v1.1**, commit `5f85fb4bc95b753a2f7d042db7dc2644ca1e8c49` (same tree as release rtl-20261101). In the repo it is the pinned submodule `external/caliptra-rtl-1x/`; see §15. Official sources used for this audit: `docs/CaliptraIntegrationSpecification.md` and `docs/CaliptraHardwareSpecification.md` at that commit. **1.x has no `strap_ss_*` / AXI DMA / Caliptra-SS straps** — those belong to 2.x / Caliptra SS.
- **Glue RTL reviewed at:** glue RTL as of `ea96100` / `ea86b5c` (unchanged since; 2FF reset synchronizers, Reset Release IP gating, warm-reset DONE-only fuse path, PAUSER lock, TIMER_CONFIG, fatal recovery, generic_input tie-off). Verilator results in §13 were built from exactly this RTL; Quartus synthesis result is from `ea86b5c`. Earlier notes: first added in `7ba4ce5` (rev 1), re-audited in `238df80` (rev 2, docs-only).
- **Getting the sources / building:** §15 (clone, submodules, `scripts/get_caliptra_a5e.sh`, `scripts/fetch_deps.sh`) and §16 (Quartus Pro 25.3 build, Docker, license, measured results, known issues), and §17 (board bring-up: pins, fit/asm/sta, programming, boot monitor and LED codes, troubleshooting).
- **Date:** 2026-11-10 (America/Chicago), revision 3 (design fixes for the rev-2 FAIL/OPEN items; checklist re-audited against the new RTL + v1.1 spec).
- **Scope:** FPGA study / production-flow *mimic*. STUDY keys, UDS, field entropy, and TRNG are **not** production secrets or a FIPS entropy source. xorshift TRNG, study UDS/FE/obf key, and the FPGA fuse-image stand-in remain **TEST-ONLY**.

This is the complete connection map and bring-up note for the Nios host design. It is based on the *actual* FPGA glue RTL, Platform Designer system, and Nios firmware, audited against the official v1.1 Integration and Hardware specifications.

Corrections vs. revision 2 (`238df80`)

Item	What changed in the design (and in this note)
M7 Reset synchronizers	`pwrgood_sync_async_assert` and `rst_sync_async_assert` (2FF) in `quartus_a5e_nios/rtl/reset_sync.sv`. Raw requests are `ctrl_pio[0/1] & sys_reset_n` (`sys_reset_n = reset_n & ~ninit_done`, see M11), then synchronized: `cptra_rst_b` async assert (to 0) / sync deassert (to 1); `cptra_pwrgood` async assert (to 1) / sync deassert (to 0), per Integration Spec Table 4. Verilator `+test=sync` **PASS**. SDC comments the 2FFs; `reset_n` remains an async false path; synchronizer *outputs* are timed to `clk`.
M10 Warm-reset fuse path	`soc_fuse_ctrl` CTRL bit 2 = `START_DONE` writes only `CPTRA_FUSE_WR_DONE=1`. Nios `caliptra_warm_reset()` holds pwrgood, pulses `rst_b`, re-locks TRNG/mailbox PAUSER (rst_b domain), then START_DONE and waits for `ready_for_runtime` (rom-1.1.1 `WarmResetFlow` resumes the existing FW; `FIRMWARE_LOAD` only if `ready_for_fw_push` asserts). Fuse PAUSER / TIMER / WDT persist (pwrgood-sticky). TB `+test=warm` **PASS**: warm `ready_for_fuses` 31 cycles after `rst_b` drop, START_DONE COUNT=1, back to `ready_for_runtime` at cycle 32,845,182 without `FIRMWARE_LOAD` (§13).
M1 `generic_input_wires`	Tied `64'b0`. Bit 0 (zeroization status) is 0; there is no zeroization source on this FPGA SoC. Bits 63:1 reserved, no flops. `magic_pio` drives only `led[7:0]`.
M12 `CPTRA_TIMER_CONFIG`	Nios / TB write `20000` (50 MHz period in picoseconds) after `ready_for_fuses`. Sticky on pwrgood; rewritten after warm reset as well (harmless).
M5 Error handling	Host polls `status_pio[5/6]` (no Qsys IRQ — see §12 open IRQ steps). Non-fatal: log and RW1C-clear `CPTRA_HW_ERROR_NON_FATAL` / `CPTRA_FW_ERROR_NON_FATAL`. Fatal: limited (`FATAL_RETRY_LIMIT=3`) `caliptra_pwrgood_cycle()` cold-reset recovery, then full cold boot. No more `fail()`-spin on the first fatal. TB `+test=fatal` **PASS** (forced fatal → pwrgood cycle → `ready_for_fuses` → full 84-word fuse push, cycle 24,065,266; §13). IRQ wiring remains **OPEN** (needs PD regen).
M2 / M3 PAUSER	After `ready_for_fuses`, firmware programs and locks **fuse**, **TRNG**, and **mailbox slot 0** to `HOST_PAUSER=1` (the value `avmm_to_apb` actually drives) before fuse START. Fuse PAUSER is pwrgood-sticky (skipped on warm reset). TRNG and mailbox locks reset on `rst_b` and are rewritten. TB `+test=pauser` **PASS**: mailbox and fuse writes from PAUSER=2 return `PSLVERR`; PAUSER=1 accepted.

Item	What changed in the design (and in this note)
M8 SRAM zeroization	Still **OPEN** for a pwrgood-time walk of TCB SRAM ports (not muxed). FPGA mitigation only: `quartus/syn_src/caliptra_sram.sv` (synthesis only) has an `initial` zero-fill, so the **mailbox SRAM** (the only `caliptra_sram` instance; 32768 × 39) is 0 in the configuration bitstream (QSF `VERILOG_CONSTANT_LOOP_LIMIT 65536` lets Quartus elaborate the loop). ICCM / DCCM / ICache use VeeR `el2_ram` and get no explicit init. This happens once at FPGA configuration — a later pwrgood cycle does **not** re-zero anything — so it is not the spec's pwrgood-time zeroization and not ASIC SRAM repair. Verilator uses the upstream `libs/rtl/caliptra_sram.sv` model, not this override.
M11 Reset Release IP / PLL	**FIXED (Reset Release); PLL N/A.** `quartus_a5e_nios/ip/rst_release.ip` = Agilex 5 **Reset Release Intel FPGA IP** (`altera_s10_user_rst_clkgate`, conduit `ninit_done`), instantiated as `u_rst_release` in `nios_caliptra_top.sv`. `sys_reset_n = reset_n & ~ninit_done` now drives the PD reset, the raw Caliptra pwrgood/rst_b requests, and a new glue 2FF (`u_sync_glue` → `glue_rst_n` for `avmm_to_apb`, `soc_fuse_ctrl`, window decode). Generated and synthesized with Quartus Pro 25.3. Verilator uses `sim/rst_release_stub.sv` (ninit_done=1 for 25 ns, like the vendor sim model). There is still no PLL (board 50 MHz direct); if one is added, AND `locked` into `sys_reset_n`.

Corrections vs. the first revision (`7ba4ce5`)

Item	What changed
`CPTRA_HW_REV_ID`	Not a flat `0x0000_0011`. It is `{SOC_STEPPING_ID[15:0], CPTRA_GENERATION[15:0]}` with generation reset value `0x0011` (major 1, minor 1, patch 0). The host reads it *before* fuses, so `0x0000_0011` is correct at that moment. RTL continuously copies `FUSE_SOC_STEPPING_ID` into the stepping field, so after the image's stepping fuse (`0x1`) is written a later read can be `0x0001_0011`.
Reset synchronizers	The v1.1 Integration Spec RDC section *requires* the integrator to add external synchronizers: `cptra_pwrgood` and `cptra_rst_b` must already be synchronized to `clk` at the Caliptra boundary. The first revision treated this as a milder CDC note on `reset_n`. Rev 2 rated it **FAIL**; fixed in rev 3 (M7).
Warm reset vs fuses	First revision correctly said the host always does a cold-style fuse push, but did not audit `soc_fuse_ctrl` against the required warm-reset path (fuses persist; SoC must still **write `FUSE_WR_DONE` again** without re-writing locked fuses). Rev 2: the IP had no done-only mode; rev 3 adds `START_DONE` (M10).
Watchdog / fatal	Added a full WDT / NMI / `cptra_error_fatal` section. The fatal *output pin* clears on `cptra_rst_b` (internal `cptra_noncore_rst_b`); sticky logs and `CPTRA_WDT_CFG` persist across warm reset; the integration requirement to *recover* from a fatal is a power-good reset. WDT t2 specifically needs a **warm reset** to restart the timers.
`CPTRA_TIMER_CONFIG`	Was omitted. SoC-writable clock period in picoseconds (sticky on pwrgood, default 0). Rev 2: Nios never programmed it; rev 3 writes 20000 (M12).
2.x names	There is no `cptra_bootfsm_bp` or `strap_ss` in 1.x. The breakpoint pin is `BootFSM_BrkPoint`. The NMI vector is an *internal register*, not a strap.

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1. Where the earlier connection-map docs live

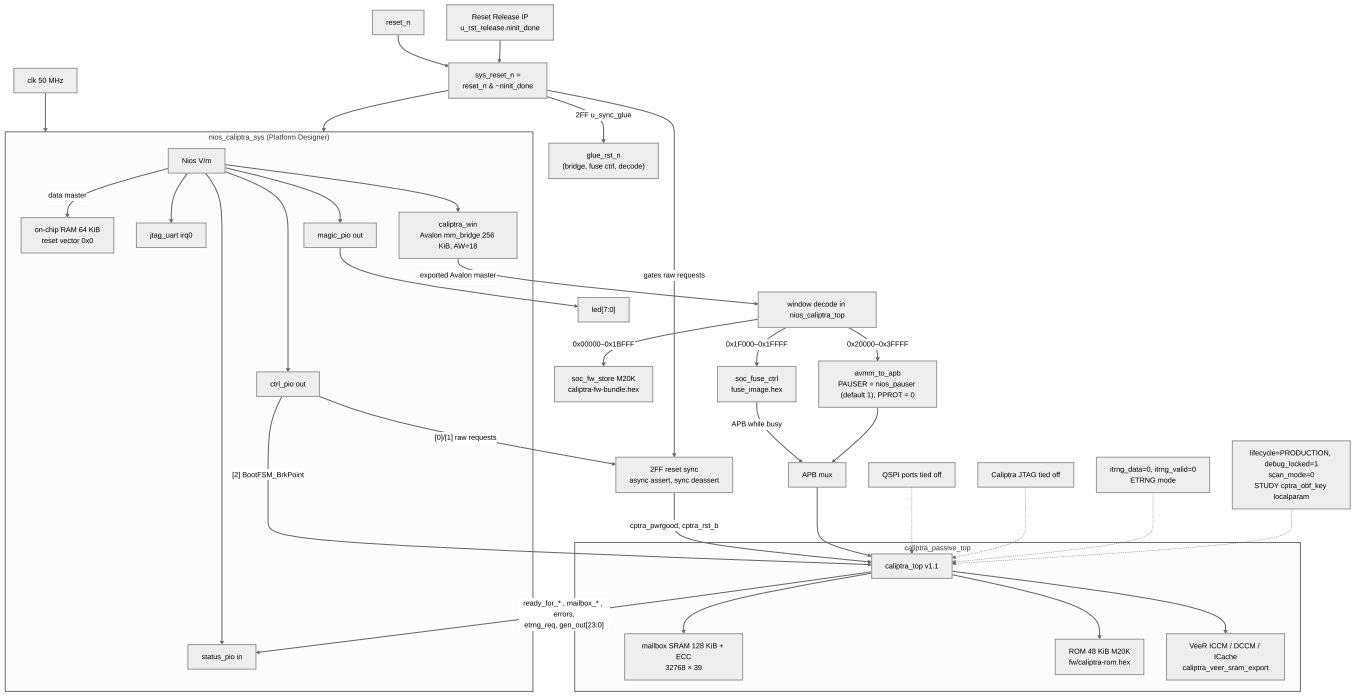
There was **no** `docs/` tree and no single integration note in the repo before the first revision of this file. The Nios ↔ Caliptra map was spread across these documents (all still present):

Document	What it covers	Status vs RTL at `32924ba`
`quartus_a5e_nios/README.md`	Closest earlier connection map: ASCII block diagram, Nios address map, straps, fuse-image summary	**Architecture / address-map / strap sections are mostly accurate.** The **Firmware** and **Not done** sections are stale (see §10).
`README.md`	Top-level overview: fuse controller, ETRNG service, FIRMWARE_LOAD, MAGIC `0xCA11F00D`	Accurate at a high level.
`sim/BOOT_NIOS_TOP_PROD.md`	Verilator production-mimic boot and the rev-3 test list (prod PASS at cycle 24,064,900, 32 TRNG fills; warm/fatal/sync/pauser PASS)	Updated for rev 3; see §13.
`STUDY.md`	C10 GX / A5E fit study (ALMs, routing OOM)	Not a signal map. Notes a Quartus critical warning for a **missing Reset Release IP** on the related A5E build. The Nios project now instantiates it (M11).

Official SoC-interface rules are in caliptra-rtl v1.1 `docs/CaliptraIntegrationSpecification.md` and `docs/CaliptraHardwareSpecification.md` at `5f85fb4`. They are in the `external/caliptra-rtl-1x/docs/` submodule after `scripts/fetch_deps.sh` (§15).

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2. Block diagram



Text equivalent (for PDF / plain renderers):

```
clk (50 MHz) + sys_reset_n = reset_n & ~ninit_done (Reset Release IP u_rst_release)
       |
       v
nios_caliptra_sys (Platform Designer)
  Nios V/m --AXI4-lite--> interconnect
       +-- ram (64 KiB OCM, reset vector 0x0, init onchip_mem.hex)
       +-- jtag_uart (stdout, IRQ 0)     +-- cpu timer @ 0x80040, debug @ 0x90000
       +-- ctrl_pio  (out) --> cptra_pwrgood / cptra_rst_b / BootFSM_BrkPoint
       +-- status_pio(in)  <-- ready_for_* / mailbox_* / errors / etrng_req / gen_out[23:0]
       +-- magic_pio (out) --> led[7:0] only (generic_input_wires tied 0)
       +-- caliptra_win (256 KiB mm_bridge, exported Avalon master, 18-bit byte address)
                   |
                   v
       window decode in nios_caliptra_top.sv
          0x00000-0x1BFFF  soc_fw_store   (M20K, fw/caliptra-fw-bundle.hex)
          0x1F000-0x1FFFF  soc_fuse_ctrl  (MMIO + APB master; fw/fuse_image.hex)
          0x20000-0x3FFFF  avmm_to_apb    (PAUSER=nios_pauser, default 1) --+
                               soc_fuse_ctrl APB (when busy)  --+--> Caliptra APB
                                                                |
                                                                v
                                       caliptra_passive_top
                                         caliptra_top v1.1
                                         mailbox SRAM, ROM, ICCM/DCCM/ICache
                                         QSPI / Caliptra JTAG / iTRNG / scan tied off
```

The Agilex 5 **Reset Release IP** (`u_rst_release`, `ninit_done`) is instantiated; `sys_reset_n = reset_n & ~ninit_done` goes to the PD clock source reset, to the glue 2FF (`glue_rst_n`), and gates the raw Caliptra requests. There is **no** PLL (or PLL-lock AND into reset) and **no** ISSP. Caliptra `cptra_pwrgood` / `cptra_rst_b` are the 2FF-synchronized versions of `ctrl_pio[0/1] & sys_reset_n`.

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3. Full signal connection table

Widths and official roles follow caliptra-rtl v1.1 `caliptra_top` / Integration Spec Tables 4–11. “FPGA source/sink” is what `nios_caliptra_top.sv` and `caliptra_passive_top.sv` actually wire.

3.1 Clocks and resets

Caliptra port	W	FPGA source / sink	Notes
`clk`	1	Board `clk` (top port). Shared by Nios, glue, and Caliptra.	SDC: 50 MHz (`period 20 ns`). **No PLL**, no on-die oscillator. Spec validates up to 400 MHz and requires an on-die oscillator for the threat model. The input clock is **ungated**; internal Caliptra clock gating is off until SoC writes `CPTRA_CLK_GATING_EN` (`0x3_00C0`, default 0, SoC-writable, resets on `cptra_rst_b`). Nios never writes it. Clock is running before Nios asserts `cptra_pwrgood`.
`cptra_pwrgood`	1	2FF `u_sync_pwrgood` of `ctrl_pio[0] & sys_reset_n`	Active-high. PIO reset value 0 ⇒ Caliptra held unpowered until Nios writes bit 0. Board `reset_n` (or `ninit_done` during device init) also takes the raw request low (**cold reset**); the pin **deasserts synchronously** (2 clocks). Assert (0→1) is asynchronous, per Table 4.
`cptra_rst_b`	1	2FF `u_sync_rst_b` of `ctrl_pio[1] & sys_reset_n`	Active-low warm-reset pin. Same raw AND, then 2FF: assert (1→0) asynchronous, deassert (0→1) synchronous. Official minimum: ≥ **10 clocks** after pwrgood before deasserting `cptra_rst_b`. Nios `delay(1000)` / TB 20 cycles meet that (and cover the 2FF).

`sys_reset_n` (= `reset_n & ~ninit_done`) also resets the Platform Designer system (clock_source `resetSynchronousEdges DEASSERT` — this sync-deasserts the **Nios/PD** reset) and, through its own 2FF `u_sync_glue` (`glue_rst_n`, async assert / sync deassert), the glue (`avmm_to_apb`, `soc_fuse_ctrl`, window-decode flops, `nios_pauser`). Caliptra's two pins use the dedicated 2FFs, not the PD synchronizer. SDC: `set_false_path -from [get_ports reset_n]` and `set_false_path -from [get_keepers {*u_rst_release*}]` (async/quasi-static sources into the synchronizers' async pins and the PD); the synchronizer outputs are registers timed to `clk`.

See **§5** for cold vs warm reset, fuse persistence, and the sequence diagram.

3.2 APB (SoC mailbox / soc_ifc)

APB is 32-bit address, 32-bit data, 32-bit PAUSER (`CALIPTRA_APB_*_WIDTH` in `config_defines.svh`). Nios sees only an 18-bit window; `avmm_to_apb` zero-extends and forces 32-bit alignment.

Caliptra port	W	FPGA source / sink	Notes
`PADDR`	32	Mux: `soc_fuse_ctrl` while `busy`, else `avmm_to_apb`	Fuse controller uses `0x0003_0200 + 4*idx` then `0x0003_00B0`. Nios path uses the 18-bit Avalon byte address.
`PPROT`	3	`avmm_to_apb` constant `3'b000`	Fuse controller does **not** drive `PPROT`; the bridge constant remains during fuse push.
`PSEL`, `PENABLE`, `PWRITE`, `PWDATA`	1 / 1 / 1 / 32	Same mux	Standard APB3 SETUP then ACCESS until `PREADY`.
`PAUSER`	32	`avmm_to_apb` `pauser_i` = `nios_pauser` (default `32'h1`); `32'h1` while `soc_fuse_ctrl` owns the mux	This is the Nios AXI/APB user ID actually used. Default mailbox valid PAUSER in RTL is `CPTRA_DEF_MBOX_VALID_PAUSER = 0xFFFFFFFF`, so Nios programs and locks mailbox slot 0, TRNG, and fuse VALID/LOCK to 1 after `ready_for_fuses` (see §4.3 / M2/M3). The TB may force `nios_pauser` to a wrong ID to check `PSLVERR`.
`PREADY`, `PRDATA`, `PSLVERR`	1 / 32 / 1	Caliptra → bridge and fuse controller	Bridge reports `avs_response = 2'b10` on `PSLVERR` (mm_bridge `USE_RESPONSE` is not enabled, so the Nios CPU does not see AXI/Avalon error responses).

3.3 Boot / mailbox notification and errors

Caliptra port	W	FPGA source / sink	Notes
`ready_for_fuses`	1	`status_pio[0]` and `soc_fuse_ctrl.ready_for_fuses`	Asserted in Boot FSM `BOOT_FUSE`. START is rejected if this is low.
`ready_for_fw_push`	1	`status_pio[1]`	Nios waits, then issues mailbox `FIRMWARE_LOAD`. Wired and used correctly.

Caliptra port	W	FPGA source / sink	Notes
`ready_for_runtime`	1	`status_pio[2]`	Success criterion; MAGIC `0xCA11F00D`. Wired and used correctly.
`mailbox_data_avail`	1	`status_pio[3]`	Level status reflecting the mailbox register. **Not** connected to a Nios IRQ. Polling is architecturally allowed (SoC is responsible for noticing data).
`mailbox_flow_done`	1	`status_pio[4]`	Same.
`cptra_error_fatal`	1	`status_pio[5]`	**Polled** (no Qsys IRQ in this PD; see open IRQ steps in §12). On fatal, `main.c` logs `CPTRA_HW_ERROR_FATAL` / `CPTRA_FW_ERROR_FATAL` and runs `caliptra_pwrgood_cycle()` (drop both pins, wait, re-release) up to three times, then a full cold boot. The pin itself is sticky until `cptra_rst_b` / `cptra_noncore_rst_b`; sticky *logs* clear only on pwrgood. Spec also: WDT t2 needs a **warm reset** to restart the timers — `caliptra_warm_reset()` is available for a host-initiated warm reset (not auto-selected by error type).
`cptra_error_non_fatal`	1	`status_pio[6]`	Polled in the boot/idle loops. Host logs and RW1C-clears `CPTRA_HW_ERROR_NON_FATAL` (`0x3_0004`) and `CPTRA_FW_ERROR_NON_FATAL` (`0x3_000C`). Same missing IRQ routing.
`BootFSM_BrkPoint`	1	`ctrl_pio[2]`	1.x name is `BootFSM_BrkPoint` (there is no `cptra_bootfsm_bp` pin). Reset value 0 (do not halt Boot FSM after fuses). If set, the FSM goes `BOOT_FUSE` → `BOOT_WAIT` until SoC writes `CPTRA_BOOTFSM_GO.GO = 1` at `0x3_00B8`. Nios never sets the strap or writes GO.

The only Platform Designer IRQ is `jtag_uart` at irqNumber 0. Caliptra status is **polled** and acted on (recovery / RW1C). Wiring the error pins as Nios IRQs needs a Qsys regen (OPEN, §12). See **§6** for WDT / NMI / fatal behavior.

3.4 TRNG

`CALIPTRA_INTERNAL_TRNG` is **commented out** in upstream `config_defines.svh`, so this build is **external-source (ETRNG) mode**. The 1.x RTL still *contains* entropy_src / CSRNG, but they are not instantiated without that define.

Caliptra port	W	FPGA source / sink	Notes
`etrng_req`	1	`status_pio[7]`	ROM/FW raise this; Nios `trng_service()` in `sw/app/main.c` polls it.
`itrng_data`	4	Tied `4'h0`	Unused in ETRNG mode.
`itrng_valid`	1	Tied `1'b0`	Forces ETRNG; no physical bitstream.

There is **no** FPGA entropy IP, no ring oscillator, and no hardware LFSR. Servicing is done in **Nios C firmware**:

1. See `status_pio[7]`.
2. Write 12 words to `CPTRA_TRNG_DATA0`…`DATA11` (`0x30078`–`0x300A4`) from an **xorshift32** seeded with `0x1234ABCD`.
3. Write `CPTRA_TRNG_STATUS = 2` (`DATA_WR_DONE`).
4. Wait for `etrng_req` to drop.

The Verilator TB does the same with `$urandom`. After `ready_for_fuses` the host programs `CPTRA_TRNG_VALID_PAUSER=1` and `CPTRA_TRNG_PAUSER_LOCK=1` (rst_b domain — rewritten after a warm reset). The entropy itself remains TEST-ONLY xorshift.

This is **TEST-ONLY**. It is not NIST-compliant entropy. The Integration Spec requires a dedicated iTRNG (or, if the shared ETRNG API is used, **immutable SoC hardware gasket logic**, not mutable firmware) and recommends locking TRNG PAUSER. See §8.

3.5 Security straps, generic wires, scan, JTAG, QSPI

Caliptra port	W	FPGA source / sink	Notes
`cptra_obf_key`	256	Localparam `STUDY_PROD_MIMIC_OBF_KEY` in `nios_caliptra_top.sv`	Fixed study value, always driven. **Not** a product secret; visible in RTL/bitstream. Real silicon would ECO/PUF this and keep it off scan. Latched inside Caliptra on a power-good cycle (internal `pwrgood_toggle_hint`).
`security_state.device_lifecycle`	2	Hardwired `2'b11` (`DEVICE_PRODUCTION`)	Production-*flow mimic*, not a live SoC lifecycle strap. Captured inside Caliptra on the first cycle after internal non-core reset release (default latched value before capture is also PRODUCTION + debug locked).
`security_state.debug_locked`	1	Hardwired `1'b1`	Debug locked (HW spec: MSB of security_state). Because of this, rom-1.1.1 arms WDT1 from `CPTRA_WDT_CFG`. JTAG would not open.
`scan_mode`	1	Tied `1'b0`	Correct for this FPGA (no DFT). Spec: must be set to 1 *before* scan is enabled so secrets can be flushed. RDC analysis also assumes 0 for functional mode.
`generic_input_wires`	64	Tied `64'b0`	Integration Spec Table 12: bit 0 = fuse zeroization status (tied 0; this FPGA SoC has no zeroization source); bits 63:1 reserved, tied 0, no storage. `magic_pio` is **not** connected here; it only drives `led[7:0]`.
`generic_output_wires`	64	`[23:0]` packed into `status_pio[31:8]`; `[63:24]` not visible to Nios	ROM/FW character log in the TB uses bits `[8:0]` (bit 8 toggles, `[7:0]` is the char).
`jtag_tck/tms/tdi`	1	Tied 0	Caliptra TAP not brought out.
`jtag_trst_n`	1	Tied 0	Held in reset. Spec wants a TAP endpoint and `jtag_trst_n` deassert synchronized to `jtag_tck`. Acceptable for a debug-locked FPGA study; not a production TAP.
`jtag_tdo`, `jtag_tdoEn`	1	Unconnected	—
`qspi_clk_o`, `qspi_cs_no`, `qspi_d_o`, `qspi_d_en_o`	—	Unconnected in the wrapper	`CALIPTRA_INTERNAL_QSPI` is not enabled.
`qspi_d_i`	4	Tied `'0`	—
UART `uart_tx`/`uart_rx`	—	Not present	`CALIPTRA_INTERNAL_UART` undefined; ports compiled out.

Nios has its **own** JTAG UART and debug module (Intel SLD / `altera_reserved_tck` in the SDC). That is unrelated to the Caliptra TAP.

Firmware image source is **`soc_fw_store`** (on-chip M20K, `fw/caliptra-fw-bundle.hex`, 109,000 bytes), a stand-in for SoC boot flash. There is no QSPI/flash controller in this design.

3.6 Memories (instantiated in `caliptra_passive_top`)

Memory	Size / shape	Implementation	Notes
Mailbox SRAM	128 KiB + 7-bit ECC, 32768 × 39	`caliptra_sram` → M20K	Export from `caliptra_top`.
BootROM (imem)	48 KiB, 6144 × 64	`caliptra_rom_m20k`, `$readmemh("../fw/caliptra-rom.hex")`	caliptra-sw **rom-1.1.1**. Do not use a 2.x ROM. Path is relative to `quartus_a5e_nios/`.
VeeR ICCM	128 KiB + ECC, 4 × 8192 × 39	`caliptra_veer_sram_export`	Error injection tied to 0.
VeeR DCCM	128 KiB + ECC, 4 × 8192 × 39	same	same
ICache (data+tag)	~16 KiB	same export	Spec notes I-Cache may be disabled/removed for synthesis; it is still instantiated.

Memory	Size / shape	Implementation	Notes
SoC FW store	28,672 × 32 (112 KiB)	`soc_fw_store` M20K	Not a Caliptra memory.
SoC fuse image	83 × 32	`$readmemh` array in `soc_fuse_ctrl`	FPGA stand-in for OTP; not non-volatile.
Nios OCM	64 KiB	PD `intel_onchip_memory`, `onchip_mem.hex`	Host code/data.

Spec also wants SRAM repair only on power-good, and SRAM contents **zeroed before `cptra_rst_b` deassert**. A true pwrgood-time walk would have to mux mailbox / ICCM / DCCM / ICache ports inside the TCB; that is **not** implemented (M8 OPEN). FPGA mitigation only: `quartus/syn_src/caliptra_sram.sv` (synthesis only) has an `initial` zero-fill, so the **mailbox SRAM** (the only `caliptra_sram` instance; 32768 × 39) is 0 in the configuration bitstream (QSF `VERILOG_CONSTANT_LOOP_LIMIT 65536` lets Quartus elaborate the loop). ICCM / DCCM / ICache use VeeR `el2_ram` and get no explicit init. This happens once at FPGA configuration — a later pwrgood cycle does **not** re-zero anything — so it is not the spec's pwrgood-time zeroization and not ASIC SRAM repair. Verilator uses the upstream `libs/rtl/caliptra_sram.sv` model, not this override.

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4. Memory map as seen by Nios

From `qsys/nios_caliptra_sys.tcl` and generated `sw/bsp/system.h`. Data-master view:

Base	Span	Block
`0x0000_0000`	64 KiB	`ram` on-chip memory (code + data, init `onchip_mem.hex`)
`0x0004_0000`	256 KiB	`caliptra_win` window (offsets below)
`0x0008_0000`	8 B	`jtag_uart`
`0x0008_0010`	16 B	`ctrl_pio` — bit 0 `cptra_pwrgood`, bit 1 `cptra_rst_b`, bit 2 `BootFSM_BrkPoint` (reset 0)
`0x0008_0020`	16 B	`status_pio` — see bit table below
`0x0008_0030`	16 B	`magic_pio` → `led[7:0]` only (not a Caliptra generic input)
`0x0008_0040`	64 B	Nios V `timer_sw_agent` (mtime) — **Nios** timer, unrelated to Caliptra `CPTRA_TIMER_CONFIG`
`0x0009_0000`	64 KiB	Nios V debug module

4.1 Window at `0x0004_0000` (18-bit offset)

Decode in `nios_caliptra_top.sv`: `address[17]` selects Caliptra; `address[16:12] == 0x1F` selects the fuse controller; everything else in the low half is the FW store.

Window offset	Nios address	Block
`0x0_0000`–`0x1_BFFF`	`0x0004_0000`–`0x0005_BFFF`	`soc_fw_store` (word-addressed firmware bundle, little-endian 32-bit)
`0x1_C000`–`0x1_EFFF`	—	**Hole:** decode still hits `soc_fw_store`, address clamped to word 0. Do not use.
`0x1_F000`–`0x1_FFFF`	`0x0005_F000`	`soc_fuse_ctrl` MMIO (4 KiB)
`0x2_0000`–`0x2_FFFF`	`0x0006_0000`	Caliptra `MBOX_CSR` (via APB)
`0x3_0000`–`0x3_FFFF`	`0x0007_0000`	Caliptra `soc_ifc` (via APB)

4.2 `soc_fuse_ctrl` MMIO (base `0x0005_F000`)

Offset	Access	Meaning
`0x00` ID	RO	`0x4655_5345` (`"FUSE"`)
`0x04` CTRL	WO	bit 0 START (83 fuse words + DONE), bit 1 CLEAR (clears DONE/ERR), bit 2 **START_DONE** (write only `CPTRA_FUSE_WR_DONE=1`, warm-reset path)
`0x08` STATUS	RO	bit 0 BUSY, bit 1 DONE, bit 2 ERR, bit 3 live `ready_for_fuses`
`0x0C` COUNT	RO	APB writes completed (84 = 83 fuse words + `FUSE_WR_DONE` on START; **1** on START_DONE)
`0x10` FW_BYTES	RO	`109000` — size of the bundle in `soc_fw_store`

Offset	Access	Meaning
0x14 NUM_FUSES	RO	83

UDS / field entropy in the on-chip image are **not** readable through this MMIO (Nios can only START and read status). After programming, Caliptra itself reads back FUSE_UDS_SEED / FUSE_FIELD_ENTROPY as 0 (write-only to the SoC), which the TB checks.

4.3 Selected Caliptra APB offsets (add to 0x0004_0000)

Offset	Register	Use in this host / notes
0x2_0000	MBOX_LOCK	Acquire before FIRMWARE_LOAD
0x2_0004	MBOX_USER	Must equal 1 (our PAUSER)
0x2_0008	MBOX_CMD	0x46574C44 ("FWLD")
0x2_000C	MBOX_DLEN	Bundle length in bytes
0x2_0010	MBOX_DATAIN	Word stream from soc_fw_store
0x2_0018	MBOX_EXECUTE	1 to start, 0 to release
0x2_001C	MBOX_STATUS	2 = CMD_COMPLETE, 3 = CMD_FAILURE
0x3_0000	CPTRA_HW_ERROR_FATAL	Sticky on pwrgood. Bit 0 ICCM ECC, bit 1 DCCM ECC, **bit 2 nmi_pin (0x4) = WDT t2**, bit 3 crypto. Writing 1 clears a bit in the *register* but **does not** deassert the cptra_error_fatal pin. Host logs this on fatal, then does a pwrgood cycle.
0x3_0004 / 0x3_0008 / 0x3_000C	CPTRA_HW_ERROR_NON_FATAL / FW_ERROR_FATAL / FW_ERROR_NON_FATAL	Non-fatal pair is RW1C-cleared by the host when status_pio[6] is set.
0x3_003C	CPTRA_FLOW_STATUS	Includes ready_for_* bits
0x3_0048 / 0x3_005C	CPTRA_MBOX_VALID_PAUSER[0] / LOCK[0]	Programmed to 1 / 1 after ready_for_fuses. Reset on cptra_rst_b — rewritten after warm reset.
0x3_0070 / 0x3_0074	CPTRA_TRNG_VALID_PAUSER / LOCK	Programmed to 1 / 1 after ready_for_fuses. Reset on cptra_rst_b — rewritten after warm reset. Default VALID reset value is 0xFFFFFFFF.
0x3_0078 – 0x3_00A4	CPTRA_TRNG_DATA[0..11]	12 ETRNG words (writes accepted only from the locked PAUSER once LOCK=1)
0x3_00AC	CPTRA_TRNG_STATUS	Write 2 = DATA_WR_DONE
0x3_00B0	CPTRA_FUSE_WR_DONE	Written by soc_fuse_ctrl, then checked by Nios. Sticky on **pwrgood**.
0x3_00B4	CPTRA_TIMER_CONFIG	Clock period in **picoseconds** (SoC RW, sticky on pwrgood, default 0). Used by Caliptra FW get_cycle_count() for MTIME. Nios / TB write **20000** (50 MHz).
0x3_00B8	CPTRA_BOOTFSM_GO	Write GO=1 to leave BOOT_WAIT if BootFSM_BrkPoint was set. Unused (strap is 0).
0x3_00C0	CPTRA_CLK_GATING_EN	SoC-writable global enable for *internal* clock gating (default 0, resets on cptra_rst_b). Nios leaves it 0 (gating disabled). (HW spec table lists 0x300300bc; the v1.1 register header / RDL address is 0x300300c0.)
0x3_00D4	CPTRA_HW_REV_ID	{SOC_STEPPING_ID, CPTRA_GENERATION}. Generation reset value 0x0011. Stepping is driven from FUSE_SOC_STEPPING_ID. Host reads this *before* fuses ⇒ 0x0000_0011. After stepping fuse 0x1 is written, a later read can be 0x0001_0011.
0x3_00D8 / 0x3_00DC	CPTRA_FW_REV_ID0/1	Read after ready_for_runtime
0x3_00E4 … 0x3_0104	CPTRA_WDT_TIMER* / STATUS	Caliptra-internal WDT programming and t1/t2 status. SoC is RO on the period/status. See §6.
0x3_0108 / 0x3_010C	CPTRA_FUSE_VALID_PAUSER / LOCK	Programmed to 1 / 1 on cold boot only (sticky on **pwrgood**; skipped after warm reset).

Offset	Register	Use in this host / notes
`0x3_0110` / `0x3_0114`	`CPTRA_WDT_CFG0/1`	SoC-provided 64-bit WDT1 timeout in cycles, sticky on **pwrgood**, default 0. Nios writes `100_000_000` / `0` after `ready_for_fuses`, before START.
`0x3_0200`–`0x3_0348`	`FUSE_*`	Written by `soc_fuse_ctrl`, not by Nios

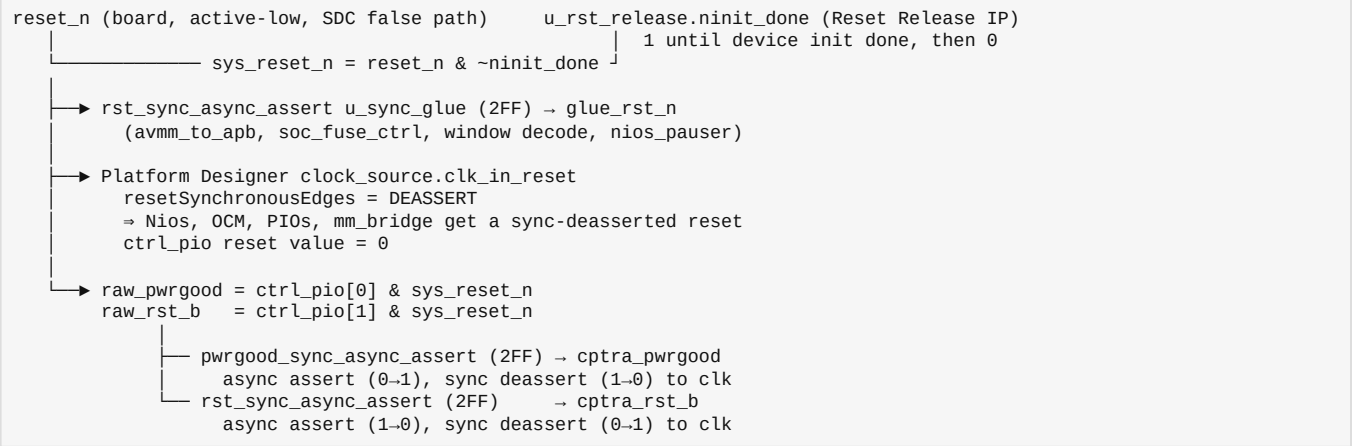
4.4 `status_pio` bits

Bit	Signal
0	`ready_for_fuses`
1	`ready_for_fw_push`
2	`ready_for_runtime`
3	`mailbox_data_avail`
4	`mailbox_flow_done`
5	`cptra_error_fatal`
6	`cptra_error_non_fatal`
7	`etrng_req`
31:8	`generic_output_wires[23:0]`

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5. Reset and power-good

5.1 How the pins are generated (this FPGA)



Present / not present for a real A5E kit bring-up:

- **Intel Reset Release IP / `ninit_done`: implemented** (`ip/rst_release.ip`, `u_rst_release`). Agilex 5 fabric is not guaranteed ready until `ninit_done` falls; until then `sys_reset_n` holds the PD, glue, and Caliptra requests in reset (M11).
- **PLL and PLL-lock.** The SDC assumes a 50 MHz input already at the `clk` port. There is no PLL-lock term in the reset qualifier.
- **ISSP** (In-System Sources and Probes) for debug-overriding resets: not instantiated.
- **External 2FF synchronizers** on `cptra_pwrgood` and `cptra_rst_b`: **implemented** in `rtl/reset_sync.sv` (`u_sync_pwrgood`, `u_sync_rst_b`). Verilator `+test=sync` checks async assert and 2- or 3-cycle sync deassert.

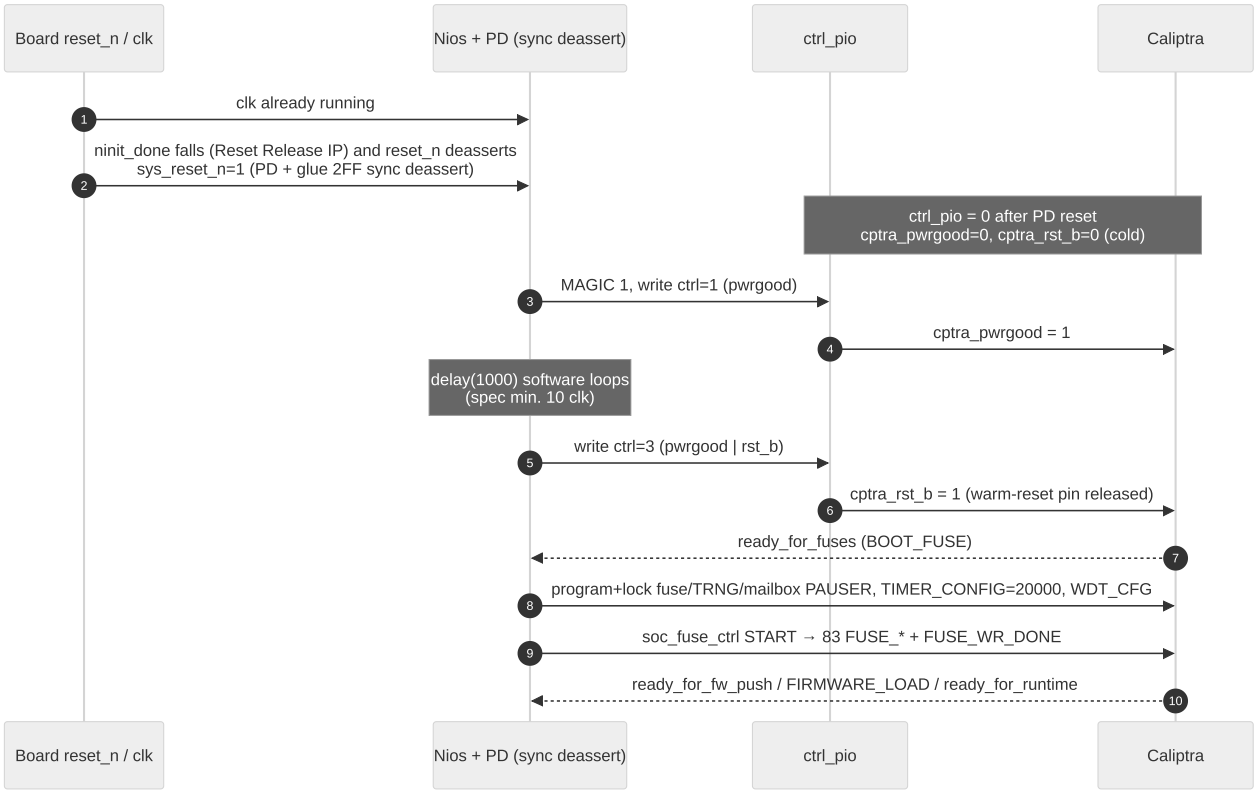
5.2 Official ordering (v1.1 Integration Spec)

- Clock must be running **before** `cptra_pwrgood` asserts. (Met: free-running board/`clk`.)
- ≥ **10 clocks** after `cptra_pwrgood` asserts before `cptra_rst_b` is deasserted. (Met in `main.c` / TB for the cold-boot path.)

- Assertions of `cptra_pwrgood` / `cptra_rst_b` may be asynchronous; **deassertions must be synchronous to `clk`**. The integrator **must add external reset synchronizers**; Caliptra assumes the pins are already synchronized at its boundary (RDC "Reset constraints and assumptions").
- **`cptra_pwrgood` deasserted** = power cycle / **cold reset**. All resettable flops return to default, including sticky registers (fuses, `FUSE_WR_DONE`, `CPTRA_WDT_CFG`, error logs, `CPTRA_TIMER_CONFIG`, fuse PAUSER lock).
- **`cptra_rst_b` deasserted while pwrgood stays high** = **warm reset**. Resets all but sticky registers. Fuses and `FUSE_WR_DONE` **persist**. The SoC must still perform a write to `FUSE_WR_DONE` while the Boot FSM is in `BOOT_FUSE` (`fuse_wr_done_observed` is cleared on `cptra_noncore_rst_b` and must see a fresh `swmod` on the done register).
- `BootFSM_BrkPoint=1` stops the FSM after fuse download until `CPTRA_BOOTFSM_GO`.

Internal Boot FSM (v1.1 `soc_ifc_boot_fsm.sv`): `BOOT_IDLE` (held while `cptra_rst_b` is asserted) → `BOOT_FUSE` (`ready_for_fuses`, non-core reset already released so the APB fuse window works) → `BOOT_DONE` (or `BOOT_WAIT` if the breakpoint strap is set). `BOOT_DONE` deasserts `cptra_uc_rst_b` through a 2FF synchronizer. A firmware-update reset only pulses the VeeR reset (`BOOT_FW_RST`).

5.3 Sequence (intended cold boot — what `main.c` actually does)



A **board `reset_n` assertion** takes both raw requests low *and* resets Nios / `ctrl_pio`, so it is always a **Caliptra cold reset** (`rst_b` falls asynchronously; `pwrgood` falls after 2 clocks). A Nios write of `ctrl_pio = 1` (pwrgood held) then back to `3` is a **warm reset**; `caliptra_warm_reset()` in `main.c` and the TB `+test=warm` path do exactly that.

5.4 How Nios and Caliptra resets are related

Event	Nios / PD	Caliptra
Board `reset_n` low	Reset (async assert, sync deassert in PD)	`rst_b` async to 0; `pwrgood` sync-deasserts after 2 clocks (cold)
Nios comes out of reset	Runs `main`, `ctrl_pio` still 0	Still held in cold reset until software says otherwise
`main` writes pwrgood, then rst_b	Running	Official cold-boot sequence (10-cycle gap plus 2FF deassert)

Event	Nios / PD	Caliptra
Nios writes `ctrl_pio = 1` then `3` (`caliptra_warm_reset`)	Running	Warm reset; fuses persist; START_DONE, then ROM resumes existing FW → `ready_for_runtime` (no FIRMWARE_LOAD)
WDT t2 / `cptra_error_fatal`	Polled; `caliptra_pwrgood_cycle()` up to 3 times, then cold re-boot	Fatal pin clears on the pwrgood/rst_b cycle; sticky logs cleared by pwrgood. `caliptra_warm_reset()` is available if only a warm reset is wanted (WDT t2).

5.5 `soc_fuse_ctrl` and warm reset

`fuse_wr_done_reg_write_observed` is cleared on `cptra_noncore_rst_b`. Advance from `BOOT_FUSE` requires `fuse_done && fuse_wr_done_observed`. After a warm reset, `CPTRA_FUSE_WR_DONE` is still 1 (sticky on pwrgood) but *observed* is 0, so the SoC must write the done register again.

`soc_fuse_ctrl` now has two start paths:

- CTRL bit 0 **START** — write all 83 `FUSE_*` words, then `FUSE_WR_DONE=1` (cold / pwrgood recovery). COUNT=84.
- CTRL bit 2 **START_DONE** — write only `CPTRA_FUSE_WR_DONE=1` (warm reset). COUNT=1.

`FUSE_WR_DONE.done.swwe` is `soc_req & ~done`, so after a warm reset (bit still 1) the write does not change the stored value. That does not matter: in the v1.1 generated register block, `hwif_out.CPTRA_FUSE_WR_DONE.done.swmod = decoded_reg_strb.CPTRA_FUSE_WR_DONE && decoded_req_is_wr` (`soc_ifc_reg.sv`), i.e. **any** APB write to `0x3_00B0` raises `swmod`, independent of `swwe`. `soc_ifc_top.sv` latches that into `fuse_wr_done_reg_write_observed` (reset on `cptra_noncore_rst_b`), and the Boot FSM leaves `BOOT_FUSE` on `fuse_done && fuse_wr_done_observed`. START_DONE issues exactly that one write, and the Verilator `+test=warm` run confirms the Boot FSM advances and the ROM reaches `ready_for_runtime` again.

After START_DONE the host re-programs TRNG and mailbox PAUSER/lock (those reset on `rst_b`). rom-1.1.1 `WarmResetFlow` then jumps to the **existing** firmware (data-vault entry point) if the previous cold boot completed; `ready_for_fw_push` / `FIRMWARE_LOAD` are not repeated. The host waits for `ready_for_runtime`. If the previous cold boot did not complete, `ready_for_fw_push` may still assert and the host loads FW as usual.

5.6 SDC / false paths

`quartus_a5e_nios/nios_caliptra.sdc`:

- `create_clock -period 20` on `clk` (50 MHz).
- `altera_reserved_tck` 40 ns, asynchronous to `clk`.
- `set_false_path -from [get_ports reset_n]` — board reset is asynchronous into the 2FF async pins and into the Nios PD.
- `set_false_path -from [get_keepers -nowarn {*u_rst_release*}]` — `ninit_done` is SDM-driven and quasi-static (falls once after configuration); it only reaches the same synchronizers. The Reset Release IP ships its own SDC for its internal clock. The **synchronized** `cptra_pwrgood` / `cptra_rst_b` are register outputs timed to `clk` and are not false-pathed onward into Caliptra.

There are no explicit constraints matching the official RDC false-path table (CPTRA_PWRGD vs other reset groups, etc.). The 2FFs provide the functional async-assert / sync-deassert the RDC analysis assumes at the Caliptra boundary.

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6. Watchdog, NMI, and fatal errors

6.1 What exists in 1.x RTL

Caliptra contains a **two-stage WDT** (`wdt.sv`), independent of VeeR, documented in the Hardware Spec “Watchdog timer” section. There are **no SoC pins** for t1/t2 timeout. Visibility to the SoC is:

- `CPTRA_WDT_STATUS.t1_timeout` / `t2_timeout` (SoC RO, `0x3_0104`)
- `CPTRA_HW_ERROR_FATAL.nmi_pin` (bit 2, mask `0x4`) on t2 in cascade mode
- The `cptra_error_fatal` output pin

Default mode is **cascade** (Timer 2 enable = 0):

1. ROM/FW enable Timer 1. Timer 1 counts until it matches its period.
2. **t1 timeout** raises an *internal* error interrupt to VeeR (`error_wdt_timer1_timeout_sts`). Timer 2 starts. If FW services t1 before t2 expires, both timers restart.
3. **t2 timeout in cascade** asserts `nmi_intr` (`nmi_intr = t2_timeout && !timer2_en`), which is connected to VeeR's NMI input, sets `CPTRA_HW_ERROR_FATAL.nmi_pin`, and sets `cptra_error_fatal`. Hardware Spec: "A warm reset is required by the SoC to reset the timers when Timer 2 times out."

If Timer 2 is enabled independently, t2 is just another error IRQ (no NMI / no fatal from this path).

Timeout periods (`CPTRA_WDT_TIMERn_TIMEOUT_PERIOD`, 64-bit, default `0xFFFF_FFFF_FFFF_FFFF`) are **Caliptra-only writable** and reset on `cptra_rst_b`. The SoC-visible knob is **`CPTRA_WDT_CFG[1:0]`** (`0x3_0110/0x114`): a 64-bit cycle count, SoC RW, sticky on **pwrgood**, default **0**. rom-1.1.1 `start_wdt()` runs only when `debug_locked=1`, reads that 64-bit value, **replaces 0 with 1 cycle**, and programs WDT1.

That is why a debug-locked boot with `WDT_CFG=0` produced `HW_ERROR_FATAL=0x4` (`nmi_pin`) at cycle 213,002, and why the host/TB now write `WDT_CFG0=100_000_000`, `WDT_CFG1=0` before fuses (~2.0 s at 50 MHz, ~250 ms at 400 MHz).

6.2 NMI vector

There is **no NMI-vector strap** in 1.x. `internal_nmi_vector` is an internal soc_ifc register; `nmi_vector` is driven to VeeR (`nmi_vec = nmi_vector[31:1]`). ROM/FW program it. The SoC does not.

6.3 SoC-side reaction in this design

- `cptra_error_fatal` / `non_fatal` are packed into `status_pio` and **polled** in `handle_errors()`. They are not Platform Designer IRQs (spec: both "shall be routed to SoC interrupt controller"). IRQ wiring is OPEN pending Qsys regen (§12).
- On **non-fatal**, the host logs and RW1C-clears `CPTRA_HW_ERROR_NON_FATAL` and `CPTRA_FW_ERROR_NON_FATAL`, then continues.
- On **fatal**, the host logs the sticky error registers and calls `caliptra_pwrgood_cycle()` (drop `ctrl_pio` to 0, delay to cover the 2FF, then cold release) up to `FATAL_RETRY_LIMIT` (3), then a full cold boot (PAUSER + TIMER + WDT + START + FIRMWARE_LOAD). It no longer `fail()`-spins on the first fatal.
- `caliptra_warm_reset()` is provided for a host-initiated warm reset (including WDT t2, which the Hardware Spec says needs a warm reset to restart the timers). The automatic fatal path uses a **pwrgood** cycle, which is the specified general fatal recovery and also clears sticky WDT_CFG / fuses (so they are reprogrammed on the following cold boot).
- `CPTRA_TIMER_CONFIG` is programmed to **20000** ps (50 MHz).

6.4 Verdict

WDT **inside** Caliptra is intact and the host's `WDT_CFG` programming is the right production-mimic fix for debug-locked ROM. The SoC now **reacts** to fatal (pwrgood recovery) and non-fatal (log/clear) via **polling**. Remaining gaps: no PD IRQ (OPEN), and the automatic path uses a cold/pwrgood cycle rather than a t2-specific warm reset (warm reset is available as `caliptra_warm_reset()`). Severity: **Medium (IRQ still OPEN; recovery path implemented and tested).**

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7. Boot / bring-up sequence

This is what `quartus_a5e_nios/sw/app/main.c` does (same sequence as `sim/tb_nios_top_fw.sv`). MAGIC values are written to `magic_pio` (LED only). Throughout, `handle_errors()` polls fatal/non-fatal: non-fatal is logged and RW1C-cleared; fatal triggers a pwrgood recovery rather than an immediate spin.

1. **MAGIC `0x1` — alive.** Nios is out of reset. Caliptra is still in cold reset (`ctrl_pio=0`).
2. **Power-good.** Write `ctrl_pio = 1` (`cptra_pwrgood`, async assert through the 2FF). Clock is already running.
3. **Wait**, then **release `cptra_rst_b`.** Write `ctrl_pio = 3` (sync deassert through the 2FF). Official Boot FSM: at least 10 clocks between pwrgood and reset release; `delay(1000)` plus a short post-deassert delay cover the 2FF.
4. **Poll `ready_for_fuses`** (`status_pio[0]`), servicing ETRNG if it appears.
5. **MAGIC `0x2`.** Read `CPTRA_HW_REV_ID` (fail if 0 or all-ones). At this pre-fuse point expect `0x0000_0011`.

6. **Program and lock PAUSER**, then timer and WDT. Order: `FUSE_VALID_PAUSER/LOCK` = 1, `TRNG_VALID_PAUSER/LOCK` = 1, `MBOX_VALID_PAUSER0/LOCK0` = 1, `CPTRA_TIMER_CONFIG` = 20000, `CPTRA_WDT_CFG` = 100,000,000 / 0. **MAGIC `0x3.`**
7. **Fuse programming (hardware IP, not Nios writing `FUSE_*`).** Check fuse-ctrl ID `0x46555345`. Write `CTRL.START` (bit 0). IP waits for `ready_for_fuses` (else ERR), then writes all **83** `FUSE_*` words from `fw/fuse_image.hex` over APB starting at `0x3_0200`, then writes `CPTRA_FUSE_WR_DONE = 1`. Nios only polls `STATUS`. **MAGIC `0x4.`**
8. **ROM boot.** Boot FSM leaves `BOOT_FUSE`, releases `cptra_uc_rst_b`, and ROM runs (including `start_wdt()` because debug is locked). Nios **must** keep servicing `etrng_req` (xorshift stand-in). Poll `ready_for_fw_push`. **MAGIC `0x5.`**
9. **Mailbox `FIRMWARE_LOAD.`** Acquire `MBOX_LOCK`, check `MBOX_USER==1`, write CMD `"FWLD"`, `DLEN=FW_BYTES` (109000), stream words from `soc_fw_store`, set `EXECUTE=1`. **MAGIC `0x6.`** Poll `MBOX_STATUS`, still servicing TRNG. Complete (2) vs failure (3). Clear `EXECUTE`. **MAGIC `0x7.`**
10. **Poll `ready_for_runtime.`** Read `CPTRA_FW_REV_ID*`.
11. **MAGIC `0xCA11F00D`** (LED = `0x0D`). Idle loop continues to service runtime TRNG requests and poll errors.

Warm reset (`caliptra_warm_reset()`, TB `+test=warm`): hold pwrgood, pulse `rst_b`, wait `ready_for_fuses`, re-lock TRNG and mailbox PAUSER (skip fuse PAUSER), `CTRL.START_DONE` (bit 2). ROM `WarmResetFlow` resumes the existing firmware; the host waits for `ready_for_runtime` (and only repeats `FIRMWARE_LOAD` if `ready_for_fw_push` asserts).

Fatal recovery: pwrgood cycle, then the cold-boot sequence from step 2, up to three retries.

Simulation of this sequence (behavioral Nios host in `sim/tb_nios_top_fw.sv`, mirroring `main.c`) is in **§13 Verification**. The full Platform Designer Nios V CPU running `main.c` has **not** been simulated.

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8. TRNG section (review)

Mode: External TRNG API (ETRNG). `CALIPTRA_INTERNAL_TRNG` is off; `itrng_valid` is tied 0, so the internal entropy_src / CSRNG path is not built, and no physical bitstream is provided.

Who services it: Nios firmware (`trng_service` in `main.c`), not a hardware gasket. Handshake is the official one: `etrng_req` → 12 words in `CPTRA_TRNG_DATA*` → `CPTRA_TRNG_STATUS.DATA_WR_DONE` → request drops. ROM/FW time out after 250,000 polls of DONE if the SoC is silent.

Entropy source: Software **xorshift32**, seed `0x1234ABCD`. Simulation uses `$urandom`. There is no FPGA TRNG, no ring-oscillator, no LFSR in RTL.

Security:

- Not a NIST SP 800-90B entropy source.
- Integration Spec: *"It is a requirement for FIPS compliance that this TRNG HW API is always handled by SoC hardware gasket logic (and not some SoC ROM or firmware code)."* This design violates that on purpose for bring-up.
- Spec also strongly prefers enabling `CALIPTRA_INTERNAL_TRNG` and wiring a physical PTRNG to `itrng_data` / `itrng_valid` (~50 kHz valid rate).
- TRNG PAUSER is now programmed to `1` and locked after `ready_for_fuses` (and again after a warm reset). API text says lock=0 means any APB agent can write TRNG DATA (and that matches this v1.1 RTL). A later "programming rules" row says unprogrammed lock means *no* agent is accepted — that row does not match this RTL. The TB confirms a write from PAUSER=2 is rejected (`PSLVERR` on mailbox/fuse; TRNG DATA writes from a wrong user are silently ignored).

Verdict: Wired correctly for an ETRNG *bring-up* hook, and enough for the ROM/FW boot that passed in Verilator. **Test-only. Do not treat boot measurements, IDevID, or any Caliptra-derived key as trustworthy on this FPGA image.**

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9. Fuse section (review)

Mechanism: A **hardware fuse-controller soft IP** (`soc_fuse_ctrl.sv`), not Nios poking `FUSE_*` registers. This matches the Integration Spec rule "SoC shall write fuse registers and fuse done via immutable logic or ROM code" *in structure* (an FSM in RTL). It is still only an FPGA stand-in for OEM OTP + sequencer: the image is a `$readmemh` of `fw/fuse_image.hex` into on-chip storage, regenerated by `scripts/gen_fuse_image.py`. Anyone who can rebuild the bitstream can change UDS / field entropy / PK hashes.

Sequencing (cold boot): Nios waits for `ready_for_fuses`, programs and locks fuse/TRNG/mailbox PAUSER plus TIMER_CONFIG and WDT_CFG, then pulses START. The IP errors if `ready_for_fuses` is low. It writes 83 words at `0x3_0200 + 4*i`, then `CPTRA_FUSE_WR_DONE=1` at `0x3_00B0`. Fuses are one-time per **power-good**. After a **warm reset**, the host pulses CTRL `START_DONE` so only `FUSE_WR_DONE` is written (see §5.5). In v1.1, `FUSE_WR_DONE.done.swmod` is any SoC write to that register (even if the bit is already 1 and `swwe` is 0), which is what `fuse_wr_done_observed` needs.

APB ownership: While `busy`, the APB mux gives the fuse IP the bus. Nios is expected only to poll fuse-ctrl STATUS (a local slave, not Caliptra APB). There is no additional hardware interlock if software also issues a Caliptra-window access during the push.

Fuse PAUSER: `CPTRA_SET_FUSE_PAUSER_INTEG = 0` in upstream `soc_ifc_pkg.sv` (lock is a programmable register, not a compile-time harden). This host writes `CPTRA_FUSE_VALID_PAUSER=1` and `CPTRA_FUSE_PAUSER_LOCK=1` after `ready_for_fuses` on a cold boot (pwrgood-sticky). Fuse-controller APB writes use PAUSER=1. A TB write from PAUSER=2 after the lock is `PSLVERR`.

Image contents (`fw/fuse_image.hex`, 83 words, STUDY values only):

Fuse	Value in this image
`FUSE_UDS_SEED[0..11]`	Deterministic STUDY words (`0x1F2E3D4C ^ 0x9E3779B9*(i+1)`). These are the words programmed into the fuse *registers* (the payload DOE consumes with `cptra_obf_key`). They are **not** documented as a real, provisioned, pre-obfuscated UDS, and they are **not** unique per device.
`FUSE_FIELD_ENTROPY[0..7]`	Deterministic STUDY words (`0xA7C3E519 ^ 0x85EBCA6B*(i+1)`). Same caveats.
`FUSE_KEY_MANIFEST_PK_HASH[0..11]`	SHA-384 of the vendor public-key descriptor in `fw/caliptra-fw-bundle.bin`, 12 big-endian words (caliptra-sw `bytes_to_be_words_48`).
`FUSE_KEY_MANIFEST_PK_HASH_MASK`	0 (no vendor ECC key revoked)
`FUSE_OWNER_PK_HASH[0..11]`	SHA-384 of the owner public-key descriptor, same packing.
`FUSE_FMC_KEY_MANIFEST_SVN`	0 (bundle FMC SVN is 0; ROM requires image SVN ≥ fuse SVN)
`FUSE_RUNTIME_SVN[0..3]`	0 (bundle RT SVN is 0)
`FUSE_ANTI_ROLLBACK_DISABLE`	0
`FUSE_IDEVID_CERT_ATTR[0..23]`	0
`FUSE_IDEVID_MANUF_HSM_ID[0..3]`	0
`FUSE_LIFE_CYCLE`	`0x3` (Production), matches the `DEVICE_PRODUCTION` strap
`FUSE_LMS_VERIFY` / `FUSE_LMS_REVOCATION`	0
`FUSE_SOC_STEPPING_ID`	`0x00000001` (A0 test); also reflected in `CPTRA_HW_REV_ID[31:16]` after it is written

Re-run `scripts/gen_fuse_image.py` if the FW bundle is re-signed. UDS/FE are never exposed through fuse-ctrl MMIO; Nios must not (and the boot ROM does not need to) read them back.

Lifecycle straps vs fuses: Both are Production (`2'b11` / fuse `0x3`) with debug locked. This is a hardwired mimic so ROM/FW take the production path (including WDT). It is not a real manufacturing/production lifecycle controller.

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10. Review findings

Severities: **High** = security or correctness risk if this image were treated as more than a study; **Medium** = real mismatch or missing integration item that should be fixed before hardware bring-up / any hardening; **Low** = limitation or tidy-up.

High — test-only / insecure if mistaken for production

ID	Finding
H1	**TRNG is a software xorshift** serviced by mutable Nios firmware over ETRNG registers. No iTRNG, no physical entropy, no hardware gasket. Official FIPS/threat-model requirements are not met.
H2	**`cptra_obf_key` is a fixed study localparam** in RTL (and therefore in the bitstream and this repo). Not PUF/HSM provisioned, not protected from scan/debug as the spec requires.

ID	Finding
H3	**Fuse “OTP” is an on-chip hex image.** STUDY UDS and field entropy are deterministic and rebuildable. No manufacturing burn, no field-entropy API with anti-replay, no debug-path disable in PRODUCTION beyond what Caliptra already does for those two register files.
H4	**Lifecycle is hardwired** to `DEVICE_PRODUCTION` + `debug_locked`. The SoC does not drive a real security-state strap.

Medium — mismatches, bugs, or missing required connections

ID	Finding
M1	**FIXED.** `generic_input_wires` is tied `64'b0`. Bit 0 (zeroization status) is 0; bits 63:1 reserved, no flops. `magic_pio` drives only `led[7:0]`.
M2	**FIXED.** After `ready_for_fuses`, host writes `CPTRA_TRNG_VALID_PAUSER=1` and `LOCK=1`. Rewritten after warm reset (rst_b domain). Entropy is still TEST-ONLY xorshift (H1).
M3	**FIXED (programmable lock).** `CPTRA_SET_FUSE_PAUSER_INTEG` is still 0 in upstream RTL. Host now programs `FUSE_VALID_PAUSER=1` and `LOCK=1` on cold boot (pwrgood-sticky). Wrong-user fuse writes `PSLVERR` in the TB.
M4	**FIXED in this revision.** `quartus_a5e_nios/README.md` Firmware / Not-done / architecture sections match the current RTL (2FF, PAUSER, START_DONE, TIMER_CONFIG, error recovery).
M5	**PARTIALLY FIXED.** Host polls `status_pio[5/6]`: non-fatal is logged and RW1C-cleared; fatal triggers a limited pwrgood-cycle recovery and cold re-boot (TB `+test=fatal` PASS through recovery to `ready_for_fuses` and a full fuse push; the rest of the cold boot is the `+test=prod` path). IRQ routing to a Nios interrupt is still **OPEN** (needs Qsys regen; steps in §12). Automatic path uses pwrgood (specified general fatal recovery); `caliptra_warm_reset()` is available for a t2-style warm reset.
M6	**No hardware mutex** beyond the `fuse_busy` APB mux. Current FW is careful; a buggy host could overlap Nios APB with the fuse FSM.
M7	**FIXED.** 2FF `u_sync_pwrgood` / `u_sync_rst_b` provide async assert and sync deassert to `clk`. Verilator `+test=sync` PASS. SDC: `reset_n` (and `u_rst_release`) are async false paths into the 2FF pins and the PD; synchronizer outputs are timed to `clk`.
M8	**OPEN**, with FPGA mitigation. A pwrgood-time zeroization FSM would mux TCB-owned SRAM ports (mailbox, ICCM, DCCM, ICache); that mux is not added. `quartus/syn_src/caliptra_sram.sv` (synthesis only) has an `initial` zero-fill, so the **mailbox SRAM** (the only `caliptra_sram` instance; 32768 × 39) is 0 in the configuration bitstream (QSF `VERILOG_CONSTANT_LOOP_LIMIT 65536` lets Quartus elaborate the loop). ICCM / DCCM / ICache use VeeR `el2_ram` and get no explicit init. This happens once at FPGA configuration — a later pwrgood cycle does **not** re-zero anything — so it is not the spec's pwrgood-time zeroization and not ASIC SRAM repair. Verilator uses the upstream `libs/rtl/caliptra_sram.sv` model, not this override.
M9	**Clock / pins:** 50 MHz board clock, no PLL, no on-die oscillator, almost all I/O still **virtual pins** (only `clk` and `reset_n` are real in the QSF). Not board-proven.
M10	**FIXED.** CTRL bit 2 `START_DONE` writes only `FUSE_WR_DONE`. `caliptra_warm_reset()` and TB `+test=warm` hold pwrgood, pulse `rst_b`, re-lock rst_b-domain PAUSER, START_DONE, then wait for `ready_for_runtime` (ROM jumps to existing FW; no `FIRMWARE_LOAD` after a completed cold boot). v1.1 `swmod` fires on any write to that register even if the bit is already 1. `+test=warm` PASS (§13).
M11	**FIXED (Reset Release IP); PLL not needed for the 50 MHz study clock.** `ip/rst_release.ip` (`altera_s10_user_rst_clkgate`, Agilex 5, conduit) → `u_rst_release.ninit_done`; `sys_reset_n = reset_n & ~ninit_done` feeds the PD, `u_sync_glue`, and the raw Caliptra requests. Quartus Pro 25.3 `quartus_ipgenerate` + `quartus_syn` include it. Board-level PLL-lock gating stays a board item (M9).
M12	**FIXED.** Host / TB write `CPTRA_TIMER_CONFIG = 20000` after `ready_for_fuses` (exercised in every boot test, §13).

Low

ID	Finding
L1	Caliptra QSPI and UART are tied off / compiled out. Expected; firmware comes from `soc_fw_store`, not flash.
L2	Caliptra JTAG TAP is tied off (`trst_n=0`). Nios JTAG UART/debug is separate and is connected. Fine for debug-locked study; not a TAP endpoint.
L3	`generic_output_wires[63:24]` never reach Nios.
L4	Window hole `0x1C000–0x1EFFF` aliases FW-store word 0.
L5	ICache SRAMs are still in the VeeR export (area leftover; not a functional bug for this 1.x config).
L6	`PSLVERR` is not propagated as an Avalon/AXI response to the Nios CPU (`USE_RESPONSE` off). Fuse IP *does* latch `PSLVERR` into its ERR bit.
L7	Full Platform Designer Nios V + interconnect is not simulated; Verilator uses a behavioral host.

Doc vs RTL mismatches (summary)

- `quartus_a5e_nios/README.md` Firmware / Not-done / architecture: updated in this revision to match RTL (M4).
- Top-level `README.md` and `sim/BOOT_NIOS_TOP_PROD.md`: updated for the new tests and the 2FF / START_DONE / PAUSER / error-recovery path.
- Revision 2 of this note recorded M1/M2/M3/M5/M7/M10/M12 as FAIL or OPEN. Those design items are fixed or narrowed as in the table above; M11 is now fixed (Reset Release IP); M8 remains OPEN; H1–H4 and the TEST-ONLY TRNG/fuse/key items are unchanged on purpose.
- Official Integration Spec vs this FPGA glue (remaining): ETRNG-by-firmware, study key, soft OTP, no iTRNG, no Caliptra TAP, virtual pins, no SRAM pwrgood walk, no error IRQ — all called out above.

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11. Spec compliance checklist

Requirements are from caliptra-rtl v1.1 Integration Spec (interface tables, Boot FSM, Fuses, RDC, Table 17 “SoC integration requirements”, error-recovery flow) and Hardware Spec (WDT, clock gating, security state), audited against the FPGA glue in the commit that lands this revision-3 text.

Requirement	Spec ref	Our implementation	Status
Clock running before `cptra_pwrgood`	Integ Table 17, clocks	Free-running `clk` (50 MHz SDC)	**PASS** (study clock; see oscillator row)
≥ 10 clocks from pwrgood to `rst_b` deassert	Integ Boot FSM	`delay(1000)` / TB 20 cycles	**PASS** (cold-boot FW path)
`cptra_pwrgood` / `cptra_rst_b` pre-synchronized to `clk`; deassert synchronous	Integ RDC constraints / Table 4	2FF `u_sync_pwrgood` (async assert 0→1, sync deassert) and `u_sync_rst_b` (async assert 1→0, sync deassert). `+test=sync` PASS	**PASS**
pwrgood deassert = cold reset; `rst_b` only = warm reset	Integ Interface rules	Wiring and firmware implement both. Board `reset_n` is a Caliptra cold reset (and resets Nios). `caliptra_warm_reset()` / `+test=warm` pulse only `rst_b`.	**PASS**
Fuses persist across warm reset; SoC must rewrite `FUSE_WR_DONE` in `BOOT_FUSE`	Integ Fuses / Boot FSM	RTL sticky as specified. `START_DONE` writes only DONE; v1.1 `swmod` fires on that write even if the bit is already 1. `+test=warm` PASS (back to `ready_for_runtime`).	**PASS**
Fuse writes + done via immutable logic or ROM	Integ Table 17	`soc_fuse_ctrl` FSM (good structure) but image is a rebuildable hex in M20K	**TEST-ONLY**
UDS/FE come from OTP, no debug inject	Integ Table 17	Hex image + STUDY values	**TEST-ONLY**
Security state driven from SoC security state	Integ Table 17 / HW Security state	Hardwired PRODUCTION + debug_locked	**TEST-ONLY**
`scan_mode` 0 in functional mode; 1 before scan	Integ DFT / RDC	Tied 0; no scan	**PASS** (FPGA functional)
`BootFSM_BrkPoint` / `CPTRA_BOOTFSM_GO`	Integ Interface rules	Strap wired to `ctrl_pio[2]`, default 0; GO unused	**PASS**
`ready_for_fuses` / `fw_push` / `runtime` connected and used	Integ Table 7	`status_pio[2:0]`, polled by FW	**PASS**
`mailbox_data_avail` / `mailbox_flow_done` visible to SoC	Integ Table 7	`status_pio[4:3]`, polled (no IRQ)	**PASS** (connectivity)
`cptra_error_fatal` / `non_fatal` routed to SoC interrupt controller	Integ error recovery / Table 17	Visible on `status_pio[5/6]` and **polled** with recovery / RW1C. Not a PD IRQ (only `jtag_uart` is).	**OPEN** (IRQ; polling + recovery implemented)
Recover fatal via power-good reset; WDT t2 via warm reset	Integ Table 17; HW WDT	`caliptra_pwrgood_cycle()` + cold re-boot (tested). `caliptra_warm_reset()` for a host-initiated warm reset (including t2).	**PASS** (pwrgood recovery tested; warm API)
SoC programs `CPTRA_WDT_CFG` before debug-locked ROM starts WDT	HW WDT; rom-1.1.1 `start_wdt`	Writes 100,000,000 / 0 before fuses	**PASS**

Requirement	Spec ref	Our implementation	Status
t1/t2 timeout visible / handled	HW WDT	No t1/t2 pins (none in 1.x). t1 is an *internal* Caliptra interrupt serviced by Caliptra FW (no SoC action required). t2 (cascade) shows up as `cptra_error_fatal` + `HW_ERROR_FATAL.nmi_pin`, which the host polls and recovers with a pwrgood cycle; `CPTRA_WDT_STATUS` is SoC-readable.	**PASS**
NMI vector strap	(n/a in 1.x)	Internal register only; correctly not a pin	**PASS** (N/A as strap)
Input clock from on-die oscillator; defend clock attacks	Integ Table 17	Board oscillator, 50 MHz, no PLL	**OPEN** (no on-die oscillator / clock-attack defence; study clock is TEST-ONLY)
SoC shall not independently power-gate Caliptra; clock gating is the `CPTRA_CLK_GATING_EN` + FW halt mechanism	Integ Table 17; HW Clock gating	Input clk ungated; `CLK_GATING_EN` left 0	**PASS**
Program `CPTRA_TIMER_CONFIG` (ps period) if MTIME conversion is used	soc_ifc RDL / drivers	Written 20000 (50 MHz) after `ready_for_fuses`	**PASS**
TRNG: dedicated iTRNG, or ETRNG via immutable gasket; lock TRNG PAUSER	Integ TRNG API / Table 17	ETRNG + Nios xorshift (TEST-ONLY, not FIPS). TRNG PAUSER now programmed and locked to 1.	**TEST-ONLY** (entropy; PAUSER lock itself PASS)
Mailbox PAUSER programmed/locked or set at integration	Integ MAILBOX PAUSER rules	Slot 0 = 1, locked in FW before mailbox use. Default remains `0xFFFFFFFF`.	**PASS** (FW path)
Fuse PAUSER locked or hardened at integration	Integ FUSE PAUSER rules	`FUSE_VALID_PAUSER`/`LOCK` written to 1 on cold boot. `CPTRA_SET_FUSE_PAUSER_INTEG` still 0 (programmable, not compile-time).	**PASS**
PAUSER driven correctly on APB	Integ Table 17	`nios_pauser` default 1; fuse mux uses 1. Host programs locks to that same ID.	**PASS**
SRAM 1-cycle read, sized for SECDED	Integ SRAM / Table 17	`caliptra_sram` / VeeR export, 39-bit mailbox	**PASS** (FPGA model)
SRAM zero + repair on pwrgood, before `rst_b` deassert	Integ Table 17	No TCB port-mux sweep. Mailbox SRAM is zero in the bitstream only (config-time, not per pwrgood); ICCM/DCCM not initialized.	**OPEN**
`generic_input` bit 0 = zeroization status; 63:1 tied 0, no flops	Integ late-binding Table 12	Tied `64'b0`; `magic_pio` is LED-only	**PASS**
Obfuscation key stable before reset, not a software value, off scan	Integ straps / Table 17	Fixed study localparam in RTL	**TEST-ONLY**
Caliptra TAP as TAP endpoint; `jtag_trst_n` sync deassert	Integ Table 17 / JTAG	TAP tied off (`trst_n=0`)	**OPEN** (no TAP endpoint; tie-off is TEST-ONLY)
QSPI / UART only if defines enabled	Integ parameters	Defines off; QSPI tied off	**PASS**
No 2.x `strap_ss_*` required	1.x `caliptra_top` ports	None present; none wired	**PASS** (N/A)
Agilex `ninit_done` / Reset Release before user logic	Intel A5E / Quartus warning	`u_rst_release` (Reset Release IP) → `sys_reset_n = reset_n & ~ninit_done` into PD, glue 2FF, and Caliptra request gating; synthesized in Quartus Pro 25.3	**PASS**

12. Open items

Still OPEN from the checklist

- **M8 — SRAM zero + repair on pwrgood, before `rst_b` deassert.** Not implemented as a walk of TCB SRAM ports (mailbox, ICCM, DCCM, ICache are owned by `caliptra_top` / the VeeR export; muxing them from SoC glue would be a TCB-adjacent change). FPGA mitigation only: `quartus/syn_src/caliptra_sram.sv` (synthesis only) has an `initial` zero-

fill, so the **mailbox SRAM** (the only `caliptra_sram` instance; 32768 × 39) is 0 in the configuration bitstream (QSF `VERILOG_CONSTANT_LOOP_LIMIT 65536` lets Quartus elaborate the loop). ICCM / DCCM / ICache use VeeR `el2_ram` and get no explicit init. This happens once at FPGA configuration — a later pwrgood cycle does **not** re-zero anything — so it is not the spec's pwrgood-time zeroization and not ASIC SRAM repair. Verilator uses the upstream `libs/rtl/caliptra_sram.sv` model, not this override. ASIC-style repair is not applicable to M20K. A real fix is a pwrgood-triggered zeroization FSM muxed onto the mailbox / ICCM / DCCM ports ahead of `cptra_rst_b` release (glue would hold `ctrl_pio[1]` low until it finishes).

- **Error IRQs (remaining part of M5).** `cptra_error_fatal` / `cptra_error_non_fatal` (and optionally `etrng_req`, `mailbox_data_avail`) are not Platform Designer IRQs. To add them: in Platform Designer, add PIO (or irq) conduits for those status bits, connect them to the Nios `irq` interface with unique irqNumbers, regenerate the BSP (`niosv-bsp`) so `system.h` has the IRQ macros, and service them in `main.c` in addition to the existing poll/recovery path.
- **Input clock source.** Board 50 MHz oscillator straight in, no PLL, no on-die oscillator or clock-glitch/frequency monitoring. If a PLL is added, AND its `locked` into `sys_reset_n`.
- **Caliptra TAP.** JTAG is tied off (`jtag_trst_n=0`). Making it a real TAP endpoint needs board pins (or the SDM JTAG chain) and a synchronized `jtag_trst_n` deassert.

Still TEST-ONLY (intentional, not closed by this revision)

- Replace ETRNG xorshift with a **hardware entropy source** (preferably `CALIPTRA_INTERNAL_TRNG` + `itrng_*`, or an immutable ETRNG gasket). TRNG PAUSER is already locked.
- Replace `soc_fuse_ctrl`'s hex image with a real (or better FPGA-emulated) **OTP** and provision unique UDS/FE. Fuse PAUSER is already locked.
- Replace the study `cptra_obf_key` localparam with a protected / HSM-provisioned source; keep it off scan.
- Drive `security_state` from a real SoC lifecycle controller rather than a hardwired PRODUCTION + debug_locked mimic.

Board / tool items

- Finish Quartus **route + timing** on a ≥32 GB host (placement already succeeded on an earlier build; routing OOM'd at ~16 GB). Programming-file / `.sof` still depends on a valid **SW-AGILEX-5E** license that includes programming-file generation (Error 23714 was seen on a related project license).
- Assign a real **board pinout** and I/O standards.
- Simulate the **real PD Nios V** system (Questa + Quartus libs), and bring the design up on the A5E kit once a SOF exists.
- Manufacturing items not in scope here: IDevID certificate provisioning, field-entropy API, FIPS zeroization API, scandump exclusion for the obfuscation key.

Checklist summary (revision 3)

Counted from the table in §11: **32** requirement rows, one primary status per row (when a cell mentions two, the worse one is primary: FAIL > OPEN > TEST-ONLY > PASS).

Primary status	Count	Rows
PASS	23	Clock before pwrgood; ≥10 clocks pwrgood→rst_b; 2FF pwrgood/rst_b sync; cold vs warm wiring; warm-reset `FUSE_WR_DONE` rewrite; `scan_mode`; BootFSM_BrkPoint; ready_for_* ; mailbox flags; fatal pwrgood recovery + warm API; `WDT_CFG` before ROM; t1/t2 visibility + recovery; NMI vector N/A; clock gating; `TIMER_CONFIG`; mailbox PAUSER lock; fuse PAUSER lock; PAUSER value on APB; SRAM shape; `generic_input_wires`; QSPI/UART tie-off; no 2.x straps; Agilex Reset Release IP.
OPEN	4	Error pins as a Nios **IRQ** (M5 remainder); input clock from an on-die oscillator / clock-attack defence; SRAM zero + repair on pwrgood (M8); Caliptra TAP endpoint (tied off).
TEST-ONLY	5	Fuse writes from a rebuildable hex image (soft OTP); STUDY UDS/FE; hardwired security state; ETRNG xorshift entropy (TRNG PAUSER lock itself PASS); study `cptra_obf_key` localparam.
FAIL	0	Every rev-2 **FAIL** (2FF sync, warm-reset fuse path, `generic_input_wires`, error handling, PAUSER) is fixed and covered by a passing Verilator test (§13).

13. Verification

13.1 Simulation (Verilator 5.054, behavioral Nios host)

> **Compiler note.** The testbench uses Verilator `--timing`, which needs C++20 coroutines. With GCC 10–12 the build fails in `verilated_timing.h` (`coroutine header requires -fcoroutines`). `sim/run_nios_top_fw.sh` calls `scripts/verilator_cflags.sh`, which adds `-CFLAGS -fcoroutines` automatically for GCC 10–12. If Verilator uses a different compiler than `${CXX:-c++}`, force it with `VERILATOR_CFLAGS=-fcoroutines sim/run_nios_top_fw.sh`, or use GCC 13 or newer.

Testbench `sim/tb_nios_top_fw.sv` drives the real `nios_caliptra_top` glue (Reset Release stub, 2FF synchronizers, `avmm_to_apb`, `soc_fuse_ctrl`, `soc_fw_store`, window decode) plus the full caliptra-rtl v1.1 core, ROM `fw/caliptra-rom.hex` (rom-1.1.1) and the signed 1.x bundle `fw/caliptra-fw-bundle.hex` (109,000 bytes). The Nios V CPU is replaced by tasks that issue the same Avalon accesses as `main.c` (same register order, PAUSER/TIMER/WDT values, fuse START/START_DONE, ETRNG service, FIRMWARE_LOAD). Clock is 100 MHz in the TB (cycle counts are clock-independent). Runs were launched 2026-11-10 10:17 America/Chicago (CST) from the glue RTL named in the header (unchanged since `ea96100`).

Test	What it checks	Result	Key cycles	Wall	Log (`sim/nios_top_fw_<x>.log`)
`+test=sync`	`cptra_pwrgood` async assert / sync deassert after 2 clocks; `cptra_rst_b` sync deassert after 2 clocks / async assert	**PASS**	done at 305 ns	<1 s	`sync`
`+test=pauser`	After fuse/TRNG/mailbox PAUSER = 1 + lock: mailbox write and fuse write with PAUSER=2 → `PSLVERR`; PAUSER=1 read of `HW_REV_ID` (`0x00000011`) accepted	**PASS**	done at ~1 µs	<1 s	`pauser`
`+test=prod`	Production cold boot: `ready_for_fuses` → HW_REV_ID `0x00000011`, `generic_input_wires == 0` → PAUSER/TIMER/WDT → full fuse push (COUNT=84, `FUSE_WR_DONE=1`) → ROM (debug-locked, WDT armed) → `ready_for_fw_push` → FIRMWARE_LOAD → `MBOX_STATUS=2` → `ready_for_runtime`	**PASS**	fuses done 355; `ready_for_fw_push` 3,449,128 (6 TRNG); FIRMWARE_LOAD issued 3,640,409; `MBOX_STATUS=2` 14,362,328; `ready_for_runtime` **24,064,900** (32 TRNG)	2,668 s	`prod`
`+test=warm`	prod, then hold pwrgood and pulse `rst_b` only; re-lock TRNG/mailbox PAUSER; `soc_fuse_ctrl` START_DONE (COUNT=1); ROM `WarmResetFlow` resumes the existing FW	**PASS**	warm `ready_for_fuses` 24,064,931 (31 cycles); `FUSE_WR_DONE` still 1 before rewrite; START_DONE done 24,064,988; `ready_for_runtime` again **32,845,182** (49 TRNG total), no `ready_for_fw_push`/FIRMWARE_LOAD	3,550 s	`warm`

Test	What it checks	Result	Key cycles	Wall	Log (`sim/nios_top_fw_<x>.log`)
`+test=fatal`	prod, then force `cptra_error_fatal`; host sees `status_pio[5]`, drops `ctrl_pio` (pwrgood cycle), cold re-release; fuses must be pushed again in full	**PASS**	fatal seen 24,064,904; `ready_for_fuses` after pwrgood cycle 24,064,956; full fuse push COUNT=84, `FUSE_WR_DONE=1` at **24,065,266**	2,682 s	`fatal`
`+test=all`	sync → pauser → prod → warm → fatal in one run	**PASS** (`TB_ALL PASS`)	prod 24,064,936; warm 32,845,218; fatal 32,845,584 (49 TRNG)	3,542 s	`all`

All six runs exit 0. Cycle counts are the TB's `cyc` counter (posedge `clk` since time 0).

Limits of this verification:

- The **fatal** test forces the `cptra_error_fatal` output pin; it does not create a real Caliptra-internal fatal (e.g. a WDT t2 NMI). It proves the SoC reaction path (detect → pwrgood cycle → cold re-boot entry → full fuse push), not Caliptra's error generation. It stops after the fuse push; the rest of the re-boot is the same path as `+test=prod`.
- The behavioral host is not the Platform Designer Nios V/m running `main.c`; PD interconnect and BSP timing are not simulated.
- Verilator uses `sim/rst_release_stub.sv` for the Reset Release IP and upstream `caliptra_sram` (no FPGA `initial` zero-fill).
- The prod/warm/fatal logs were produced by a TB build that also printed a trailing `TB_ALL PASS` line in single-test runs (Verilator `$finish` completes the time step). The TB now prints `TB_ALL` only for `+test=all`; the sync/pauser logs were re-run with that build. This does not change any result.

Rerun: `sim/run_nios_top_fw.sh +test=<name>` (builds into `sim/obj_nios_top_fw/`, needs `external/caliptra-rtl-1x`: run `scripts/fetch_deps.sh` first). A full `+test=all` takes about an hour on this 15 GB box.

13.2 Synthesis (Quartus Prime Pro 25.3.0 Build 109, A5EC065BB32AE6S)

`quartus_a5e_nios/` at `ea86b5c` (with the Reset Release IP): **Successful**, logic utilization **138,715 / 222,400 ALMs (62 %)**, 145,784 dedicated logic registers, 47 DSP blocks (`quartus_a5e_nios/output_files/nios_caliptra_a5e.syn.summary`). Fit/route and timing were **not** run (needs ≥ 32 GB RAM; placement passed on an earlier build). No `.sof`: programming-file generation needs a license that includes it. Re-verified from a fresh clone with the repo-relative QSF and submodule RTL: see §16.

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14. Source index

Path	Role
`external/caliptra-rtl-1x/`	Upstream caliptra-rtl v1.1 (`5f85fb4`), git submodule, unmodified
`quartus/syn_src/`	Synthesis overrides: `caliptra_sram.sv`, `caliptra_top_tb_pkg.sv`, `caliptra_veer_sram_export.sv`
`quartus_a5e_nios/nios_caliptra_a5e.qsf` / `.qpf`	Nios project (repo-relative paths into `../external/caliptra-rtl-1x`, `../rtl`, `../quartus/syn_src`, `../fw`)
`scripts/get_caliptra_a5e.sh`, `scripts/fetch_deps.sh`, `scripts/quartus_docker.sh`	Clone/update, pin check, Quartus-in-Docker wrapper (§15, §16)
`rtl/caliptra_passive_top.sv`	Caliptra 1.x wrapper, memories, QSPI tie-offs
`quartus_a5e_nios/rtl/nios_caliptra_top.sv`	Top: Reset Release gating, PD system, window decode, APB mux, 2FF reset requests + glue reset, generic_input tie-off, straps
`quartus_a5e_nios/rtl/avmm_to_apb.sv`	Avalon → APB3, `pauser_i` (default 1)
`quartus_a5e_nios/rtl/soc_fuse_ctrl.sv`	Fuse sequencer soft IP (START + START_DONE)

Path	Role
`quartus_a5e_nios/rtl/reset_sync.sv`	2FF `pwrgood_sync_async_assert` / `rst_sync_async_assert`
`quartus_a5e_nios/rtl/soc_fw_store.sv`	FW-bundle M20K
`quartus_a5e_nios/qsys/nios_caliptra_sys.tcl`	Regeneratable PD system
`quartus_a5e_nios/sw/app/main.c`	Nios host: cold/warm boot, PAUSER locks, TIMER_CONFIG, WDT_CFG, fuse START/START_DONE, ETRNG, FW load, fatal/non-fatal handling
`quartus_a5e_nios/nios_caliptra.sdc`	50 MHz `clk`, false paths on async `reset_n` and `u_rst_release` (2FF outputs timed to clk)
`quartus_a5e_nios/ip/rst_release.ip`	Agilex 5 Reset Release IP (`ninit_done`)
`sim/rst_release_stub.sv`	Verilator stand-in for the Reset Release IP
`scripts/gen_fuse_image.py` / `fw/fuse_image.hex`	STUDY OTP image
`fw/caliptra-rom.hex`, `fw/caliptra-fw-bundle.hex`	ROM 1.1.1 and signed 1.x bundle
`sim/tb_nios_top_fw.sv`, `sim/run_nios_top_fw.sh`	Verilator TB: `+test=prod|sync|pauser|warm|fatal|all`
`sim/nios_top_fw_{sync,pauser,prod,warm,fatal,all}.log`	Rev-3 passing logs (§13)
caliptra-rtl v1.1 `5f85fb4` `docs/CaliptraIntegrationSpecification.md`	Official SoC interface, Boot FSM, RDC, Table 17
caliptra-rtl v1.1 `docs/CaliptraHardwareSpecification.md`	WDT, clock gating, security state
caliptra-rtl v1.1 `src/soc_ifc/rtl/soc_ifc_boot_fsm.sv`, `wdt.sv`, `soc_ifc_top.sv`	Boot FSM, WDT/NMI/fatal, fuse sticky / observed

15. Getting the sources

The upstream Caliptra RTL is **not copied into this repo**. It comes in as pinned git submodules under `external/`, used unmodified. Every FPGA-specific change lives in this repo.

Path	What	Version / commit
`external/caliptra-rtl-1x/`	chipsalliance/caliptra-rtl (submodule)	**v1.1**, `5f85fb4bc95b753a2f7d042db7dc2644ca1e8c49`
`external/caliptra-rtl-2x/`	chipsalliance/caliptra-rtl (submodule)	**v2.1.2**, `49370266d12cb0c4a8f71b3a0ff7e54ba7d4866e`
`external/caliptra-rtl-2x/submodules/adams-bridge/`	chipsalliance/adams-bridge (nested submodule)	v2.0.3, `b77e3d899e828d626cfc2a0d26a6b5704cc121e0`
`rtl/`, `quartus/syn_src/`	1.x passive wrapper + synthesis overrides (`caliptra_sram`, `caliptra_top_tb_pkg`, `caliptra_veer_sram_export`)	this repo
`quartus_a5e_nios/`	Nios V + Caliptra 1.x Quartus project: `rtl/` glue, `qsys/` Platform Designer system (generated output committed), `ip/rst_release.ip`, `sw/` BSP + app, `onchip_mem.hex`	this repo
`quartus_a5e/`	Caliptra 1.x core-only Quartus project (size study)	this repo
`fw/`	1.x ROM (rom-1.1.1), signed FW bundle, fuse image	this repo
`sim/`	Verilator TB for the 1.x Nios glue	this repo

Path	What	Version / commit
`caliptra2x/`	2.x glue (`rtl_2x/`), sim (`sim_2x/`, `sim/`), firmware (`fw_2x/`), fuse script (`scripts/`), core-only Quartus project (`quartus_a5e_2x/`)	this repo
`scripts/get_caliptra_a5e.sh`	Clone or update this repo with all submodules, then verify the pins	this repo
`scripts/fetch_deps.sh`	`git submodule update --init --recursive` (or clone the pinned commits if there is no `.git`), then verify; `--check` verifies only	this repo
`scripts/quartus_docker.sh`	Run any Quartus command in the Quartus Pro 25.3 Docker image with the repo and license mounted	this repo

All paths in the `.qsf`, `.f` and run scripts are **relative to the file's own directory**, so the checkout can be anywhere. If Quartus reports `caliptra_passive_top` or `caliptra_top` as undefined, the submodules are missing: run `scripts/fetch_deps.sh`.

Download (macOS or Linux; needs git; about 1 GB with both RTL trees):

```
# Option A: one script (clones, or updates an existing clone with pull + submodule update, then checks pins)
curl -fsSLO https://raw.githubusercontent.com/pwilliam282/caliptra-a5e-quartus/main/scripts/get_caliptra_a5e.sh
bash get_caliptra_a5e.sh ~/work/caliptra-a5e-quartus

# Option B: plain git
git clone --recursive https://github.com/pwilliam282/caliptra-a5e-quartus.git
cd caliptra-a5e-quartus
scripts/fetch_deps.sh            # expect: ok ... x3, "deps OK"
```

If you cloned without `--recursive`, or downloaded a ZIP, run `scripts/fetch_deps.sh` anyway.

---

16. Quartus build

Tools and host.

- Quartus Prime **Pro 25.3.0 Build 109** with Agilex 5 device support. All results in this guide come from that version.
- Quartus Pro runs only on **Linux or Windows x86-64**. On a Mac (including Apple-silicon MacBooks), use the Mac for git and editing, and run Quartus on a Linux x86-64 PC, VM or server. The Docker image is `linux/amd64` and is not usable under emulation on Apple-silicon Docker Desktop.
- Docker option: the official image `alterafpga/quartuspro-v25.3:agilex5`, wrapped by `scripts/quartus_docker.sh <dir> '<command>'`. The wrapper mounts the whole repo (including `external/`) at the same path, mounts your license read-only, runs as your UID, and uses `--net=host`.

License. The A5EC065 needs the no-cost **SW-AGILEX-5E** license from Altera's Self-Service Licensing Center. It is node-locked to the NIC MAC address of the machine that runs Quartus. In Docker this works because of `--net=host`: FlexLM reads the host NIC, and without it `quartus_syn` can segfault in the host-ID lookup. Point the tools at the file:

- Native: `export LM_LICENSE_FILE=/path/to/SW-AGILEX-5E.dat`
- Docker: `export LIC_HOST=/path/to/SW-AGILEX-5E.dat`, or put the file at `license/SW-AGILEX-5E.dat` in the repo (gitignored)

Never commit license `.dat` files.

Projects.

Project	Directory	`.qpf` / revision	Top
Nios V + Caliptra 1.x (the design in this guide)	`quartus_a5e_nios/`	`nios_caliptra_a5e`	`nios_caliptra_top`
Caliptra 1.x core only (size study)	`quartus_a5e/`	`caliptra_a5e`	`caliptra_passive_top`
Caliptra 2.x core only (fit gate; 126 %, does not fit)	`caliptra2x/quartus_a5e_2x/`	`caliptra2x_a5e`	`caliptra_passive_top`

Step 1: generate IP (Nios project only). The Platform Designer output (`qsys/nios_caliptra_sys/`, `qsys/ip/nios_caliptra_sys/`) is committed. The Reset Release IP output (`ip/rst_release/`) is gitignored. Run `quartus_ipgenerate` once after cloning (and after any `.qsys`/`.ip` change). It regenerates every `QSYS_FILE`/`IP_FILE` in the project:

```
cd quartus_a5e_nios
quartus_ipgenerate nios_caliptra_a5e
# Optional, only if you changed the PD system: rebuild it from Tcl first
#   cd qsys && qsys-script --script=nios_caliptra_sys.tcl --quartus-project=none && cd ..
#   (or: qsys-generate qsys/nios_caliptra_sys.qsys --synthesis=VERILOG --quartus-project=nios_caliptra_a5e)
```

`run_nios_a5e_docker.sh [syn|fit]` does the full regeneration (Tcl → PD → ipgenerate → synthesis) in one go. The Nios program memory is initialised from the committed `quartus_a5e_nios/onchip_mem.hex`. Rebuild it with `quartus_a5e_nios/sw/build_app.sh` (needs `RISCV_BIN=<dir with riscv-none-elf-gcc>`) only if you change `sw/app/main.c`.

Step 2: compile. Native Quartus on Linux:

```
cd quartus_a5e_nios
quartus_syn --analysis_and_elaboration nios_caliptra_a5e   # quick check: all files found, hierarchy elaborates
quartus_syn nios_caliptra_a5e                              # Analysis & Synthesis
quartus_fit nios_caliptra_a5e                              # place + route (needs >= 32 GB RAM)
quartus_sta nios_caliptra_a5e                              # timing
quartus_asm nios_caliptra_a5e                              # .sof (only with real pins and a license that enables it)
# or all stages:  quartus_sh --flow compile nios_caliptra_a5e
```

Same thing with Docker, from the repo root:

```
export LIC_HOST=$HOME/licenses/SW-AGILEX-5E.dat
scripts/quartus_docker.sh quartus_a5e_nios 'quartus_ipgenerate nios_caliptra_a5e && quartus_syn nios_caliptra_a5e'
scripts/quartus_docker.sh quartus_a5e_nios 'quartus_fit nios_caliptra_a5e && quartus_sta nios_caliptra_a5e'
scripts/quartus_docker.sh quartus_a5e 'quartus_syn caliptra_a5e'
scripts/quartus_docker.sh caliptra2x/quartus_a5e_2x 'quartus_syn caliptra2x_a5e'
```

GUI alternative.

1. *File → Open Project →* `quartus_a5e_nios/nios_caliptra_a5e.qpf`.
2. *Tools → Generate IP* (or accept the prompt to generate IP).
3. *Processing → Start → Start Analysis & Synthesis* (or *Start Compilation* for the full flow).

Outputs. Reports go to `<project>/output_files/`:

- `<rev>.syn.summary`: ALMs, registers, DSP
- `<rev>.syn.rpt`: hierarchy, resource usage by entity, messages
- `<rev>.syn.ae.rpt`: Analysis & Elaboration
- `<rev>.fit.*`, `<rev>.sta.*`, `<rev>.sof` after fit, timing and assembler

The `qdb/`, `dni/` and `tmp-clearbox/` work directories are gitignored.

Measured results (Quartus Pro 25.3.0 b109, A5EC065BB32AE6S, from a fresh `git clone --recursive` of commit `28951ad` on a 16 GB / 8-core Linux box, Docker image above):

Project	Analysis & Elaboration	Full Analysis & Synthesis
`quartus_a5e_nios` (`nios_caliptra_top`)	**OK**, 0 errors, 1 min 19 s, 3.3 GB peak; `quartus_ipgenerate` OK, 0 errors, 9 s	**Successful**, 0 errors: **138,715 / 222,400 ALMs (62 %)**, 145,784 registers, 47 DSP post-merge, 5,671,936 block-memory bits; 6 min 50 s at `--parallel=4`, 4.0 GB peak. This matches the earlier build exactly (§13.2)
`quartus_a5e` (`caliptra_passive_top` 1.x)	**OK**, 0 errors, 44 s, 2.8 GB peak	141,256 / 222,400 ALMs (**64 %**), 142,373 registers, 47 DSP (committed `syn.summary`; 6 min 30 s, 3.6 GB)
`caliptra2x_a5e` (`caliptra_passive_top` 2.x)	**OK**, 0 errors, 1 min 41 s, 4.4 GB peak	279,685 / 222,400 ALMs (**126 %, does not fit**), 411,569 registers, 199 DSP (committed report; 22 min, 6.8 GB)

All three hierarchies elaborate with no missing-file or undefined-entity messages. The ROM, FW-bundle and fuse hex files are read by `$readmemh` from their repo-relative paths (`../fw/…`, `../fw_2x/…`).

Expected warnings (harmless):

- Critical Warning 16668: MFCU turns off auto-discovery. MFCU is intentional, because Caliptra relies on macros persisting across files.
- Critical Warning 20759 on the core-only projects: no Reset Release IP. That IP is only in the Nios top.
- Critical Warning 127004 (Nios project, ×2): Quartus pads the internal MIFs for the ROM (6144 → 8192 words) and the FW store (28672 → 32768 words) to a power of two. This is expected.
- Critical Warning 18061/18010: power-up-level notes on the security-state latch.
- Verilog truncation and parameter warnings from upstream RTL.

Known issues.

- **Intermittent Quartus start-up crash in Docker:** `Error (22912): Unhandled exception … DDM_T::verify_token(token) : Cannot identify the client` within the first second. It is not a design problem. Simply rerun the same command; on the build box it took up to 3 tries.
- **Synthesis hangs with no CPU use.** Quartus can stop progressing, usually during the SLD/JTAG-hub `ip-deploy` step or right after RAM inference when it starts its parallel helper processes. Signs: the log stops growing, `quartus_syn` sits at about 0 % CPU, and defunct `quartus_sh` children remain. Most likely the same start-up crash, hitting a helper process. Kill the container (`docker kill <id>`; with `DOCKER_NAME=<name>` set for `scripts/quartus_docker.sh` you can `docker kill <name>`) and rerun. A healthy Nios synthesis shows 4 busy `quartus_syn` helpers with `--parallel=4`. The run then finishes normally; the earlier note recorded about 7 min at `--parallel=8`.
- **Memory.** A&E needs about 3–4.5 GB, and full synthesis about 4 GB (1.x) or 7 GB (2.x). **Fit/route needs ≥ 32 GB**: placement passed on a 16 GB VM but routing ran out of memory, so fit, timing and `.sof` have not been produced yet.
- **Pins.** All top-level I/O except `clk` (and `reset_n` in the Nios project) are `VIRTUAL_PIN ON`. Assign real pins and I/O standards for your board before `quartus_asm`.

---

17. Board bring-up: boot and firmware load

This applies to the **1.x Nios project `quartus_a5e_nios/` only**. The 2.x design does not fit the A5E 065 (126 %, see the 2.x guide §14), so there is no 2.x bitstream.

17.1 No separate firmware-load step

Everything Caliptra needs is baked into the bitstream at synthesis. There is nothing to download over JTAG after configuration.

Content	Source file	Where it ends up
Caliptra ROM (rom-1.1.1)	`fw/caliptra-rom.hex`	`caliptra_rom_m20k` in `rtl/caliptra_passive_top.sv` (`$readmemh`)
FMC + runtime FW bundle (109,000 B)	`fw/caliptra-fw-bundle.hex`	`soc_fw_store` M20K (stand-in for boot flash)
Fuse image (STUDY values)	`fw/fuse_image.hex`	`soc_fuse_ctrl` image ROM
Nios program	`quartus_a5e_nios/onchip_mem.hex`	Nios on-chip RAM (`nios_caliptra_sys_ram`)

After configuration, the Reset Release IP drops `ninit_done` and `main.c` runs the whole boot on its own (§7):

1. Power-good / `rst_b` release.
2. `ready_for_fuses` → PAUSER locks, `TIMER_CONFIG`, `WDT_CFG` → `soc_fuse_ctrl` fuse push + `FUSE_WR_DONE`.
3. ROM boots while the host services `etrng_req`.
4. `ready_for_fw_push` → mailbox **FIRMWARE_LOAD** from `soc_fw_store`.
5. `ready_for_runtime`.
6. Runtime loop: TRNG service and error polling.

17.2 Pin assignment

The QSF makes every top-level port virtual except `clk` and `reset_n` (`set_instance_assignment -name VIRTUAL_PIN ON -to *`, then `OFF` for `clk` and `reset_n`). For a board:

1. Delete the three `VIRTUAL_PIN` lines in `quartus_a5e_nios/nios_caliptra_a5e.qsf`.
2. Add a location and an I/O standard for each top port, using your board's schematic. Pin names and standards below are placeholders:

```
set_location_assignment PIN_<clk_pin>    -to clk          ;# 50 MHz board oscillator (nios_caliptra.sdc: create_clock -
period 20.000)
set_location_assignment PIN_<rst_pin>    -to reset_n      ;# push-button, active low
set_location_assignment PIN_<led0_pin>   -to led[0]       ;# ... led[1] .. led[7]
set_instance_assignment -name IO_STANDARD "<bank standard, e.g. 1.2 V / 1.8 V / 3.3 V LVCMOS>" -to clk
set_instance_assignment -name IO_STANDARD "<...>" -to reset_n
set_instance_assignment -name IO_STANDARD "<...>" -to led[*]
```

Port	Dir	Notes
`clk`	in	Single clock for Nios and Caliptra. The SDC constrains **50 MHz** (20 ns). If your oscillator differs, change `nios_caliptra.sdc`, `CPTRA_TIMER_CONFIG` (20000 ps in `main.c`) and the BSP clock
`reset_n`	in	Active-low board reset, asynchronous (false-pathed in the SDC). Gated with `ninit_done` inside the top
`led[7:0]`	out	`MAGIC_PIO[7:0]`, the boot progress code (§17.5)

The **JTAG UART** and Nios debug use the device's dedicated JTAG pins through the SLD hub and need no user pins.

17.3 Fit, assemble, timing

```
cd quartus_a5e_nios
./dock.sh 'quartus_ipgenerate nios_caliptra_a5e && quartus_syn nios_caliptra_a5e'   # if not done yet (§16)
./dock.sh 'quartus_fit nios_caliptra_a5e && quartus_asm nios_caliptra_a5e && quartus_sta nios_caliptra_a5e'
```

- **Memory:** this needs ≥ **32 GB RAM**. Routing ran out of memory on a 16 GB VM (§16).
- **Output:** `quartus_a5e_nios/output_files/nios_caliptra_a5e.sof`.
- **Timing:** check the timing reports (`output_files/nios_caliptra_a5e.sta.rpt`, or the Timing Analyzer GUI) before trusting the board. Every corner must have **positive setup and hold slack on `clk` at 50 MHz**. Timing closure at 50 MHz has **not** been demonstrated yet, because no fit has run. If it fails, lower the clock (and update the SDC and `TIMER_CONFIG`).
- **License:** `quartus_asm` needs a license that enables programming-file generation for this device.

17.4 Program the FPGA

Program from the Linux/Windows machine the board's USB-Blaster is plugged into. There is no Quartus Programmer for macOS. The free standalone Quartus Prime Programmer is enough for this step and needs no license.

```
jtagconfig                      # lists cables and the JTAG chain, e.g. "1) USB-BlasterIII [1-3]" + device IDs
quartus_pgm -c 1 -m jtag -o "p;output_files/nios_caliptra_a5e.sof"
# If the chain has more than one device (e.g. on some dev kits), add the device index from jtagconfig:
#   quartus_pgm -c 1 -m jtag -o "p;output_files/nios_caliptra_a5e.sof@<index>"
```

- `-c` takes the cable number or the full cable name printed by `jtagconfig`.
- The `-o "p;file.sof[@index]"` syntax was checked against `quartus_pgm --help=o` in Quartus Pro 25.3.
- **From Docker:** run `DOCKER_USB=1 ../scripts/quartus_docker.sh "$PWD" 'jtagconfig && quartus_pgm …'`. This adds `--privileged -v /dev/bus/usb`. The host user still needs USB-Blaster udev permissions. This path has not been tested with real hardware.

17.5 Monitor the boot

```
juart-terminal -c 1             # in Quartus Pro 25.3: $QUARTUS_ROOTDIR/bin/juart-terminal (-c CABLE, -d DEVICE, -i
INSTANCE)
# then press reset_n (or reprogram) to watch the boot from the start
```

The JTAG UART prints the progress lines below (from `main.c`; the values shown are the expected ones for the committed ROM and bundle):

```
CPTRA_HW_REV_ID   = 00000011
FIRMWARE_LOAD bytes= 0001A9C8
MBOX_STATUS       = 00000002
ready_for_runtime FLOW_STATUS = ...
CPTRA_FW_REV_ID0  = ...
```

The `led[7:0]` value is `MAGIC[7:0]`:

MAGIC	led[7:0]	Meaning (main.c)
0x00000001	0x01	Nios alive; caliptra_cold_release() (pwrgood, then rst_b)
0x00000002	0x02	ready_for_fuses seen; CPTRA_HW_REV_ID read OK
0x00000003	0x03	PAUSER (fuse, TRNG, mailbox) locked; TIMER_CONFIG and WDT_CFG written
0x00000004	0x04	soc_fuse_ctrl fuse push + FUSE_WR_DONE complete; ROM running
0x00000005	0x05	ready_for_fw_push; ROM waiting for firmware
0x00000006	0x06	FIRMWARE_LOAD written to the mailbox and EXECUTE set; waiting for MBOX_STATUS
0x00000007	0x07	FIRMWARE_LOAD accepted (MBOX_STATUS = 2, CMD_COMPLETE); waiting for ready_for_runtime
0xCA11F00D	0x0D	**ready_for_runtime**: boot complete; host keeps serving TRNG
0xDEAD000E	0x0E	Fail: timeout waiting for a status bit
0xDEAD000F	0x0F	Fail: bad HW_REV_ID, fuse controller error, or FIRMWARE_LOAD MBOX_STATUS = 3 (CMD_FAILURE)
0xDEAD00FF	0xFF	Fail: more than FATAL_RETRY_LIMIT (3) fatal errors / pwrgood recoveries
0xDEAD0010	0x10	Fail: cold re-boot after a runtime fatal error did not succeed
0x13, 0x14, 0x15, 0x17	same	Warm-reset path (caliptra_warm_reset()), only if you call it

In simulation, `ready_for_runtime` comes about 24.06 M cycles after release (§13.1), so expect about 0.5 s at 50 MHz plus the Nios polling overhead.

17.6 Troubleshooting

Symptom	Likely cause / what to check
LEDs dark or stuck at 0x01	Nios runs but Caliptra never reaches ready_for_fuses. Check that reset_n is released and has the right polarity. Check that the Reset Release IP's ninit_done went low (sys_reset_n = reset_n & ~ninit_done). Check that ctrl_pio drives pwrgood/rst_b (bits 0 and 1). If the LEDs are dark, Nios is not running at all: check clk, the pins and onchip_mem.hex
Stuck at 0x04 or 0x05 (then 0x0E)	The ROM is not progressing. Read CPTRA_FW_ERROR_FATAL / CPTRA_FW_ERROR_NON_FATAL (0x3_0008 / 0x3_000C in the Caliptra window, Nios address 0x0004_0000 + offset) and CPTRA_HW_ERROR_*, for example from the Nios debugger or a debug print. Typical causes: etrng_req not served (TRNG PAUSER not locked to 1, or the host loop not running); or a fuse/bundle mismatch, where the KEY_MANIFEST_PK_HASH and owner fuses don't match fw/caliptra-fw-bundle.bin
Fails at 0x06 / 0x07 (0x0F, MBOX_STATUS = 3)	The ROM rejected the image. The bundle and the fuse image don't belong together. Regenerate the fuses from the bundle with python3 scripts/gen_fuse_image.py (writes fw/fuse_image.hex), then re-synthesize and fit. Check fw/SHA256SUMS. Also check that FW_BYTES in nios_caliptra_top.sv (109,000) matches the bundle size
0xFF / 0x10	Repeated cptra_error_fatal. Read CPTRA_HW_ERROR_FATAL / CPTRA_FW_ERROR_FATAL before the pwrgood cycle clears them (add a print in handle_errors())

17.7 Changing the Nios firmware

`main.c` lives in the Nios on-chip RAM image, which is part of the bitstream. To change it:

1. Edit `quartus_a5e_nios/sw/app/main.c`.

2. Run `RISCV_BIN=/path/to/xpack-riscv-none-elf-gcc/bin quartus_a5e_nios/sw/build_app.sh`. This rebuilds the ELF and regenerates `quartus_a5e_nios/onchip_mem.hex`.
3. Re-run synthesis, fit, asm and sta (§17.3) and reprogram.

The same rebuild applies to a new Caliptra ROM, bundle or fuse image in `fw/`, because they are `$readmemh`-initialised. (Updating only the memory-initialisation contents without a refit is possible in Quartus, but has not been set up here.)

17.8 Caveat

This flow is **not board-proven**: no fit, timing closure or hardware run has been done yet. It also uses **STUDY** values everywhere: UDS, field entropy, the `cptra_obf_key` strap, test-signed firmware, and an xorshift TRNG that is not an entropy source. Use the bitstream for **bring-up only**. It must not hold or attest anything real (§10 High items).